4EE34: EXTERNAL PROJECT

Power System Studies of Industrial Distribution System Using ETAP

A

Report

Submitted in partial fulfillment of the

4EE34: External Project

Award of B. Tech. (Electrical Engineering) Degree

By

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Birla Vishvakarma Mahavidyalaya(Engineering College)

(An Autonomous Institute)

Vallabh Vidyanagar: 388120

(May 2026)

Birla Vishvakarma Mahavidyalaya (Engineering College)

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2. Duration of Project Work : 4 Months

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10. Name of the course coordinator: Dr. Akshay A. Pandya

Industry Certificate

ELCON ENGINEERS
Private Limited
Power System Consultants
And Engineers

EEPL/IS-018/26 April 15, 2026

To,
The Head of Department
Electrical Department,
Birla Vishvakarma Mahavidyalaya Engineering College.
Vallabh Vidyanagar – 388120.

Dear Sir,

This is to certify that **Mr. Vahora Mahmmadaadil Amiraehmad** (Enrollment No. : 220070109072) has completed his internship in our company successfully from 22 December, 2025 to 15 April, 2026.

During internship period, He has attended the office regularly and was found sincere and obedient.

We wish him all the success for his future career.

Yours faithfully,

For **Elcon Engineers Private Limited**

ELCON ENGINEERS PVT. LTD. VADODARA

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Certificate

Certified that, work presented by **Mr. Mahmmadaadil Amiraehmad Vahora** bearing ID.No. **22EE074** and Enrollment No: **220070109072** in the Project entitled **Power System Studies of Industrial Electrical Systems using ETAP** for 4EE34 : External Project of B. Tech (Electrical) Final year is satisfactory to warrant its acceptance for the purpose of evaluation. It is to be understood that by this approval the undersigned does not necessarily endorse or approve any statement made, opinion expressed or conclusions drawn but approve the study for the purpose for which it is submitted.

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Head

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Acknowledgement

As part of our academic program, we are extremely proud and satisfied to be working on the significant project "Power System Studies of Industrial Distribution System using ETAP."

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Abstract

This project uses ETAP (Electrical Transient Analyzer Program) software to provide a thorough analysis and simulation of the electrical power system. The main goal is to use sophisticated computational modelling to assess the operational performance, dependability, and safety of a Mineral Plant .

To optimize voltage profiles and equipment loading, the study starts with the creation of an accurate One-Line Diagram (OLD) and a steady-state Load Flow Analysis. A Short Circuit Analysis was carried out in compliance with IEC standards to confirm the interrupting ratings of protective devices in order to guarantee the physical integrity of the infrastructure.

In order to achieve elective tripping sequences and reduce the impact of localized faults on the larger system, the project also describes the implementation of Protective Device Coordination (Star). In order to ensure compliance with NFPA 70E/IEEE 1584 safety requirements, an Arc Flash Analysis was conducted to ascertain incident energy levels and flash boundaries. The study's findings offer a technical road map for improving system stability, averting equipment failure, and creating a strict safety procedure for electrical workers.

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List of Abbreviations

Software & Methodologies

- **ETAP**: Electrical Transient Analyzer Program
- **OLD**: One-Line Diagram
- **SLD**: Single Line Diagram

System Components & Equipment

- **CLR**: Current Limiting Reactor
- **CT**: Current Transformer
- **ERMS**: Energy Reducing Maintenance Switch
- **I/C**: Incomer / Incoming
- **MCC**: Motor Control Center
- **NGR**: Neutral Grounding Resistor
- **O/G**: Outgoing
- **ONAF**: Oil Natural Air Forced
- **ONAN**: Oil Natural Air Natural
- **PDC**: Power Distribution Center
- **SEL**: Schweitzer Engineering Laboratories
- **VFD**: Variable Frequency Drive
- **WHRS**: Waste Heat Recovery System
- **XLPE**: Cross-linked Polyethylene

Protection & Relaying

- **CTI**: Coordination Time Interval
- **DD**: Device Duty
- **DTOC**: Definite Time Overcurrent
- **IDMT**: Inverse Definite Minimum Time
- **PDC**: Protective Device Coordination
- **PSM**: Plug Setting Multiplier
- **RC**: Relay Coordination
- **TCC**: Time-Current Characteristic
- **TMS**: Time Multiplier Setting
- **ZSI**: Zone Selective Interlocking

Arc Flash & Safety

- **AFB**: Arc Flash Boundary
- **AFH**: Arc Flash Hazard
- **FCT**: Fault Clearing Time
- **HCB**: Horizontal Conductors in a Box
- **HOA**: Horizontal Conductors in Open Air

- **HRG**: High-Resistance Grounding
- **IE**: Incident Energy
- **PPE**: Personal Protective Equipment
- **VCB**: Vertical Conductors in a Box
- **VCBB**: Vertical Conductors terminated in an insulating Barrier
- **VOA**: Vertical Conductors in Open Air

Electrical Parameters & Concepts

- **AC**: Alternating Current
- **DOL**: Direct-On-Line
- **LRC**: Locked Rotor Current
- **THD**: Total Harmonic Distortion

Standard Units

- **kA**: Kiloampere
- **kV**: Kilovolt
- **kvar**: Kilovolt-Ampere Reactive
- **MVA**: Megavolt-Ampere
- **MW**: Megawatt

List of Nomenclature

C	Capacitance
I	Actual fault current
I_n	Full Load Current / Nominal Current
I_s	Relay pickup current
I_{sc}	Short circuit current
I''_a	Arcing fault current
k,α,η	Curve constants for IDMT Relays
L	Inductance
P	Active Power / Real Power
ρ	Density
Q	Reactive Power
R	Resistance
R_0, X_0, L_0, C_0	Zero sequence parameters (Resistance, Reactance, Inductance, Capacitance)
S	Apparent Power
t	Fault clearing time / Operating time
V	Voltage
X	Reactance
X/R	Reactance to Resistance Ratio
Xd′′	Sub-transient reactance
Y	Admittance / Susceptance
Z	Impedance
σ	Conductivity
δt	Coordination margin
cal/cm²	Calories per square centimetre (Unit of Incident Energy)

Chapter - 01 - Project Orientation

- **Overview and Background**

One of the most power-intensive industries is mineral processing, which depends on a steady and uninterrupted supply of electricity to keep heavy grinding, crushing, and flotation circuits running. The goal of this project is to use ETAP to conduct a thorough electrical analysis of a Mineral Processing Plant

To avoid "nuisance tripping," which can result in hours of expensive downtime for mechanical clearing, the electrical network in such facilities must be built to withstand the high inrush currents of large motors while maintaining voltage stability.

- **Project Objectives (Technical Scope)**

The focus of this study is based on five engineering pillars:

1. **High-Fidelity Modeling :** Creating a digital twin of the plant's Power Control Center (PCC), Motor Control Centers (MCCs), and captive generation units.
2. **Steady-State Load Flow :** Analyzing the system under "Peak Production" and "Minimum Load" scenarios to improve transformer tap settings and power factor correction.
3. **Short Circuit Evaluation :** Calculating 3-phase and Line-to-Ground faults to make sure all circuit breakers are rated safely according to IEC 60909 standards.
4. **Star Protection & Coordination :** Developing a selective coordination scheme so that the device closest to the fault trips first. This keeps the rest of the plant up and running.
5. **Arc Flash Hazard Assessment :** Determining the Incident Energy (cal/cm^2) at each bus to define PPE requirements and ensure compliance with NFPA 70E safety standards.

The main goal of this project is to improve the efficiency, reliability of protection, and stability of the industrial power system. Using ETAP simulations and analysis, the project gives recommendations for optimizing electrical system design, updating protection schemes, and ensuring adherence to industrial safety standards.

This study enhances academic knowledge and provides practical benefits to industrial power system planners and plant maintenance teams. It shows how important proactive system analysis is for preventing failures and supporting sustainable industrial operations.

Chapter - 02 - Company Introduction



Figure 1 :- Company Logo

Company Name : Elcon Engineers Private Limited

Address : Elcon Engineers Pvt. Ltd. 501-2-3, Onyx Business Centre, Akshar Chowk, O. P. Road, Vadodara - 390020, Gujarat, India

Contact No : +91 265 2985152, +91 8401927636

Email Id : eepl@elconengineers.com

Website : https://elconengineers.com/

2.1 About Company

Elcon Engineers Pvt. Ltd., established in 1982 and headquartered in Vadodara, is a premier electrical engineering consultancy firm. The company specializes in power protection and automation of industrial electric systems, especially those with captive power plants, grid support, and distribution sub-station networks. Elcon is also a recognized system integrator for Schweitzer Engineering Laboratories (SEL), offering reliable, advanced solutions for industries across India and globally.

2.2 Vision

To be a leading and preferred service provider for electrical Power System Conceptual Design, System Study and Solutions in the world by targeting beyond Customer's expectation and satisfaction.

2.3 Mission

- To provide quality, cost effective and value-added services to Customers meeting their requirements and time schedule.
- To focus continuous growth of individual and team by acquiring knowledge, continuous learning and training.
- To build up an efficient and effective team of power system experts using state of the art IT tools, infrastructure and innovative approaches.
- To establish an encouraging working atmosphere by developing respectful, empowering, trust worthy and transparent working team.

2.4 Key Services Offered

- Power System Design Engineering Consultancy & Start-up Support
- Special Engineering Solutions
- Power System Study, Network Analysis and Solutions
- Customized Training Programs on Industrial Electric Power Systems
- Power Management Systems
- Relays Retrofit Solutions, Relay and Scheme Testing

Chapter - 03 - Project Defination , Identification and Objective

3.1 Project Defination

Industrial electrical distribution networks are essential for resource-heavy operations like mineral processing plants. Continuous power is necessary for high-inertia machinery, such as ball mills, crushers, and conveyors. These systems need to ensure high power quality and reliability, even under harsh environmental conditions and significant load changes. In a mineral plant, one "nuisance trip" or a voltage drop during a large motor start can cause mechanical blockages and lead to major financial losses from production downtime.

This project focuses on the analysis and optimization of a Mineral Processing Plant's electrical distribution system using ETAP software. The study includes modeling the Single Line Diagram (SLD), performing Load Flow Analysis, Short Circuit Analysis, and Protective Device Coordination and Arc Flash Analysis. The overall goal is to enhance system efficiency, stability, and fault resilience.

3.2 Project Identification

- **Project Title :** " Power system studies of industrial distribution system using ETAP"
- **Domain :** Electrical Power Systems, Industrial Distribution Network, Power System Protection
- **Software Tool Used :** ETAP (Electrical Transient Analyzer Program)
- **Industry Focus :** Mineral Processing Plant
- **Key Focus Areas :**
    1. Load Flow Analysis
    2. Short Circuit Analysis
    3. Relay Coordination
    4. Arc Flash Analysis

3.3 Primary Objectives

The primary objectives of this project are as follows:

1. System Modeling

To model the complete electrical network of a mineral processing plant using ETAP software. This involves preparing a detailed Single Line Diagram (SLD) that includes the utility grid, onsite distribution transformers, and high-capacity motor loads such as crushers and mills.

2. Load Flow Analysis

To perform steady-state Load Flow Analysis to :-

- Analyze how real and reactive power is distributed across the processing circuits.
- Evaluate voltage stability at all major buses during peak operational loads.
- Identify overloaded cables, transformers, or areas with poor voltage regulation.
- Calculate system losses and find chances for improving the power factor.

3. Short Circuit Analysis

To conduct Short Circuit Analysis to :-

- Determine the maximum fault current magnitudes (Isc) at various locations within the plant.
- Verify the Interrupting Rating and adequacy of circuit breakers and switchgear.
- Identify critical fault zones to ensure equipment can withstand thermal and magnetic stresses during a failure.

4. Relay Coordination Studies (Protection Star)

To carry out Relay Coordination Studies to :-

- Establish Time-Current Characteristic (TCC) curves for protective devices.
- Ensure Time-Selective fault clearance, isolating only the faulted section.
- Minimize unnecessary tripping of the main mill circuits and improve overall system selectivity.
- Increase system reliability and prevent production-stopping blackouts.

5. Arc Flash Hazard Analysis

To perform an Arc Flash Analysis to :-

- Calculate the Incident Energy levels (cal/cm^2) at each electrical bus following IEEE 1584 and NFPA 70E standards.
- Determine the Flash Protection Boundaries to ensure a safe working distance for plant electricians.

- Assign suitable Personal Protective Equipment (PPE) categories for maintenance tasks.
- Generate standardized safety labels to reduce the risk of injury from arc-blast events.

Chapter - 04 Literature Reviews

- **Research Paper- 01**

"Power System Load Flow Analysis in ETAP"

Author: Prof. Vishal V. Mehtre, Sr. Ishwar Katakwar

Published in: International Journal of Innovative Research in Electrical, Electronics, Instrumentation and Control Engineering Impact Factor 8.021 Peer-reviewed & Refereed journal Vol. 12, Edition 4, April 2024

Final Thoughts :-

In summary, this study successfully illustrates the use of ETAP software for performing load flow analysis on a 33 kV distribution network. Through the modeling of the system's single line diagram and simulation of actual operating conditions, critical performance metrics such as bus voltage levels, power flows, and equipment loading were assessed.

The findings indicate that transformer overload and imbalanced load distribution greatly influence voltage stability and overall system efficiency. Detecting these problems early allows engineers to implement corrective measures such as load balancing, placing capacitors, or adjusting tap changers to ensure system performance stays within acceptable thresholds.

The research underscores the significance of utilizing simulation tools such as ETAP for planning, optimization, and problem-solving in power systems. It aids in guaranteeing dependable and safe power distribution while also establishing a foundation for advanced studies such as contingency analysis, protection coordination, and renewable integration.

This study emphasizes the importance of load flow analyses in enhancing system reliability, reducing downtime, and facilitating informed choices in industrial power networks.

- **Research Paper - 02**

"ETAP - Based Load Flow Analysis, Short Circuit Analysis, And Relay Protection"

Authors :- Dr. S. Sankarakumar M.E., Ph.D., A. Sri Bala Krishnan M. Srinivasapandian, S. Surdish Muthu, Tamilselvan.

International Journal of Advanced Research in Core Engineering Sciences and Technology (IJARBEST), Volume 7, Issue 4, April 2021.

Final Thoughts :-

This project involved the implementation of load flow analysis, short circuit study, and relay protection using ETAP software to assess and improve the reliability and safety of the power system. The main aim was to tackle problems like under-voltage and fault-driven overcurrent by simulating real-time system performance and protective measures.

The load flow analysis assisted in assessing if system voltages stay within permissible limits during different operational conditions, such as the abrupt failure of generators, transformers, transmission lines, or significant loads. This offered understanding of the correct voltage management and load allocation throughout the network. Short circuit analysis simulated fault conditions and assessed the maximum short circuit current, determined to be 12 kA—well below the safe operating capacity of the current circuit breaker panel rated at 80 kA, providing a solid safety margin.

According to the short circuit findings, relay coordination was developed utilizing overcurrent relays. These relays successfully identified faults and transmitted tripping commands to circuit breakers, guaranteeing quick fault isolation. This verifies that the protection scheme of the system is correctly set up to protect equipment and ensure operational continuity.

In summary, the simulation based on ETAP shows that an integrated examination of load flow, short circuit occurrences, and relay protection is essential for planning, dependability, and fault tolerance in contemporary power systems

- **Research Paper - 03**

"Arc Flash Assessment of Medium Voltage Power Systems utilizing ETAP"

Author :- S.N. Chauhan, S.M. Pujara, P.K. Makhijani, C.D. Kotwal

Published in :- Released in the International Journal of Computer Sciences and Engineering Research Paper Volume 7, Issue 3, March 2019 E-ISSN: 2347-2693

Final Thoughts :-

The IEEE 30 bus network is modeled and simulated in ETAP to assess the hazard level and subsequently enhance protections for working personnel and equipment following an AFH analysis. The impact of various parameters on incident energy is examined using this simulated IEEE 30 bus system. The analysis indicates that incident energy rises with higher short circuit current, prolonged fault clearing time, and greater spacing between conductors, while it diminishes with an increase in working distance.

Of all these four parameters, fault clearing time and working distance significantly influence incident energy. Thus, greater focus needs to be placed on developing and analyzing the different methods that can minimize fault clearing time and enhance the operational distance

Chapter 05 - Mineral Processing Distribution Network Study

A Mineral Processing Distribution Network Study is about deeply looking into. Making models of electrical distribution systems in big mining and processing places. These industries usually work on voltage levels. Often 33kV or 11kV for power intake stepping down to 4.16kV or 6.6kV for big mill motors and 415V/600V for smaller pumps and conveyor belts.

Unlike manufacturing mineral plants use huge inductive loads that work all the time, 24/7. The study is about checking this network through simulations like Load Flow, Short Circuit, Relay Coordination and Arc Flash Analysis using ETAP. This is to make sure the facility can handle starting torques and keep safety in a high-risk place.

5.1 Key Objectives

1. **Production Continuity :** Stop shutdowns of primary mills to avoid costly problems.
2. **Voltage Stability :** Manage the impact of high-power motor starting on the rest of the facility.
3. **Fault Resilience :** Ensure switchgear can handle fault currents common in heavy-duty distribution.
4. **Personnel Safety :** Measure Arc Flash hazards to protect workers in high-vibration areas.
5. **Standard Compliance :** Follow IEEE 1584 and NFPA 70E for safety and operational excellence.

5.2 Common Challenges Faced by Mineral Processing Plants

1. **Voltage Sags :** Big drops during the startup of Ball Mills or Crushers causing sensitive control PLCs to reset.
2. **Coordination Complexity :** Balancing fast fault clearance with the need to ride through motor-starting inrush currents.
3. **Environmental Risks :** Accumulation of conductive ore dust increasing the probability of phase-to-phase faults and arc events.
4. **Operational Downtime :** The cost of failure where an electrical trip leads to hours of manually clearing stalled conveyor lines.

5.3 ETAP-Based Solutions for Mineral Processing Challenges

1. Load Flow Analysis

- Challenge: Keeping voltage stable while running grinding and flotation circuits at the same time.

- Solution: ETAP's Load Flow Analysis makes voltage regulation better. It. Balances the distribution of loads to ensure proper voltage levels at the Motor Control Centers (MCCs). By analyzing "Peak Production" scenarios ETAP helps find the necessary transformer tap settings and capacitor bank sizes to prevent undervoltage tripping.

2. Short Circuit Analysis

- Challenge: Verifying if heavy-duty switchgear can survive the energy released during a fault in a mineral plant.
- Solution: ETAP's Short Circuit Analysis calculates fault magnitudes based on IEEE C37 / IEC 60909. This ensures that the plant's circuit breakers are adequately rated for the "Duty" they will face, preventing equipment explosions and ensuring long-term system integrity.

3. Protection Coordination Studies

- Challenge: Ensuring that a jammed conveyor motor trips its breaker without shutting down the entire processing line.
- Solution: Using the ETAP Star module Time-Current Characteristic (TCC) curves are developed. The study optimizes the grading between relays and fuses ensuring Selective Coordination. This limits the impact of a fault to the possible zone protecting the continuous flow of the mineral process.

4. Arc Flash Hazard Analysis

- Challenge: Protecting electricians from thermal energy hazards during maintenance of high-power mill starters.
- Solution: ETAP's Arc Flash Analysis calculates the incident energy. Determines safe flash boundaries. In a mineral plant, where maintenance's frequent this data is used to generate safety labels and define required PPE Categories ensuring compliance, with NFPA 70E and reducing the risk of fatal injuries.

5. Harmonic Analysis (Optional/Supporting)

- Challenge: Managing distortion caused by Variable Frequency Drives (VFDs) used for conveyor speed control.
- Solution: ETAP identifies sources and evaluates Total Harmonic Distortion (THD). It recommends filter placements to prevent the overheating of mineral-grading electronics and distribution transformers.

Chapter-06 ETAP Modelling

6.1 Grid Modelling

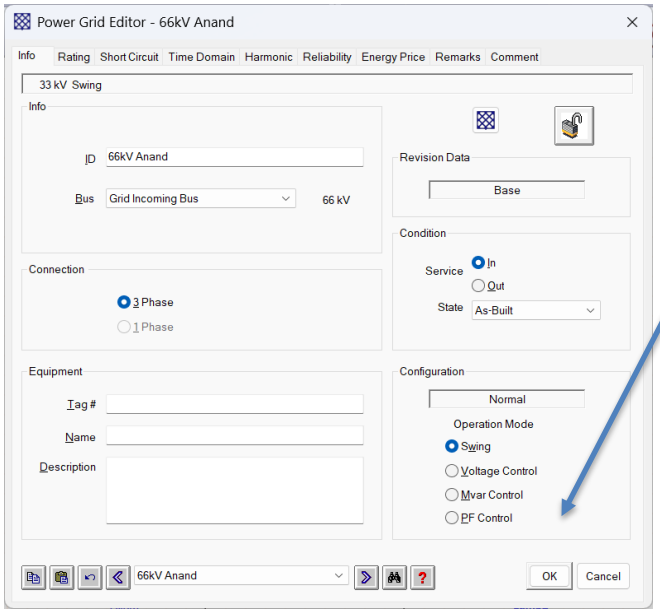


Figure 2 :- Grid Info Tab

- **Swing** : Acts as the reference bus with fixed Voltage (V) and Angle (Delta), supplying whatever P and Q are needed to balance the system.
- **Voltage Control** : Operates as a PV bus where you fix Real Power (P) and Voltage (V), while the grid adjusts Reactive Power (Q) to maintain that voltage.
- **Mvar Control** : Operates as a PQ bus where both Real Power (P) and Reactive Power (Q) are fixed, allowing the voltage to fluctuate.
- **PF Control** : Operates as a PQ bus where Real Power (P) and Power Factor (PF) are fixed, automatically calculating the required Q to maintain that ratio.

220kV Mogar	220 KV	29.06	11072.50
220kV Mogar	66 KV	32.18	3678.62
66kV Anand	66 KV	21.35	2440.49
66kV Anand AG	66 KV	20.08	2295.51
66kV Anklav	66 KV	7.19	822.37
66kV Bhalej	66 KV	5.07	579.25
66kV Bhetasi	66 KV	7.31	835.45
66kV Borsad	66 KV	5.36	613.23

Figure 3 :- Anand Fault Levels from GETCO

Typical Values Of X/R Ratio :-

- 14 for 50 Hz Supply
- 17 for 60 Hz Supply

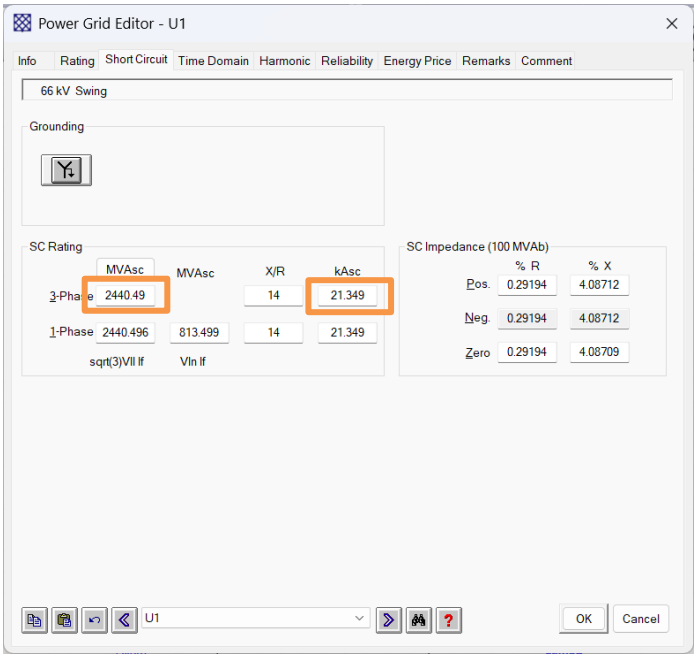


Figure 4 :- Grid Rating Tab

6.2 Generator Modelling

Rated speed	RPM	3000
Main Ratings		
Rated output	MW/MVA	600 MW , 705.9 MVA
i) Max. Continuous Rating	MW/MVA	600 MW , 705.9 MVA
ii) Temp limits		Within Class B
i) Generator rating at Turbine VWO condition	MW/MVA	630 MW , 741.2 MVA
ii) Temp limits		Within Class F
Rated power factor	--	0.85 (Lag)
Rated terminal voltage	kV	21
Rated stator current	Amps.	19407
Phase	---	3
Rated frequency	Hz	50
Excitation at rated output and power factor		
a) Voltage	Volts.	413
b) Current	Amps.	5192
a) 100% load		98.79
b) 75% load		98.84
c) 50% load		98.76
d) 25% load		98.16

Figure 5 :- Generator Datasheet

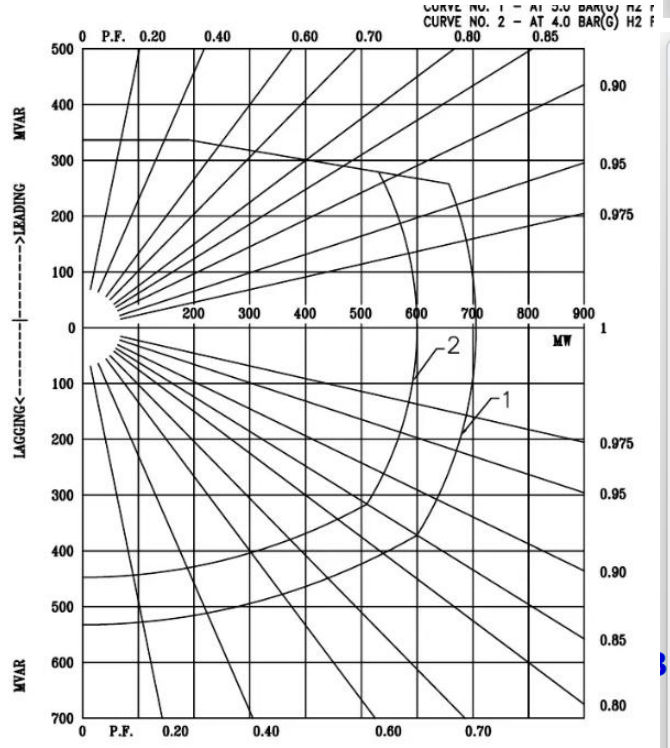


Figure 7 :- Generator Capability Curve Graph

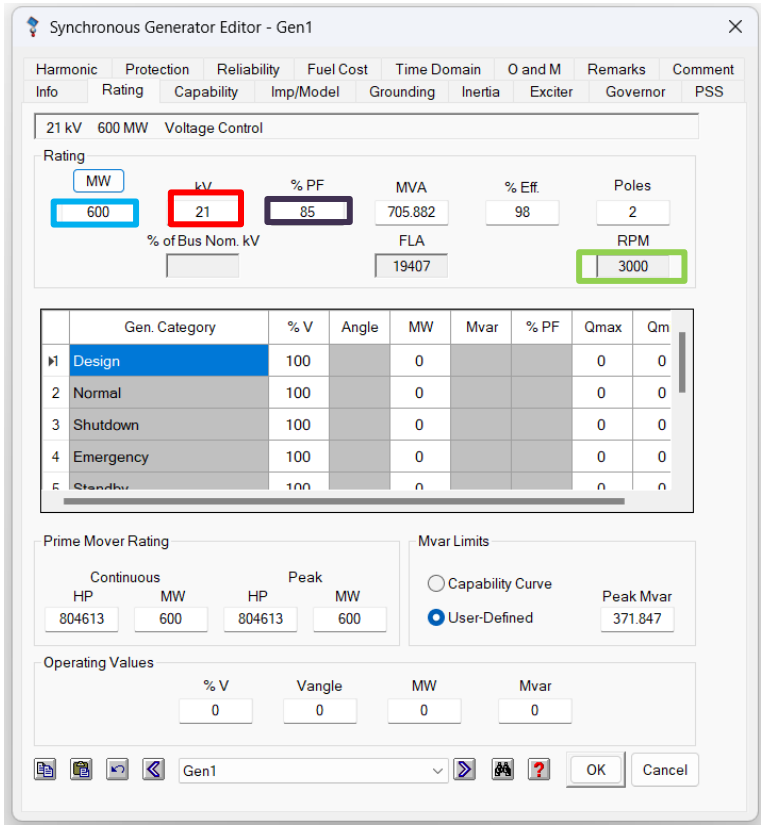
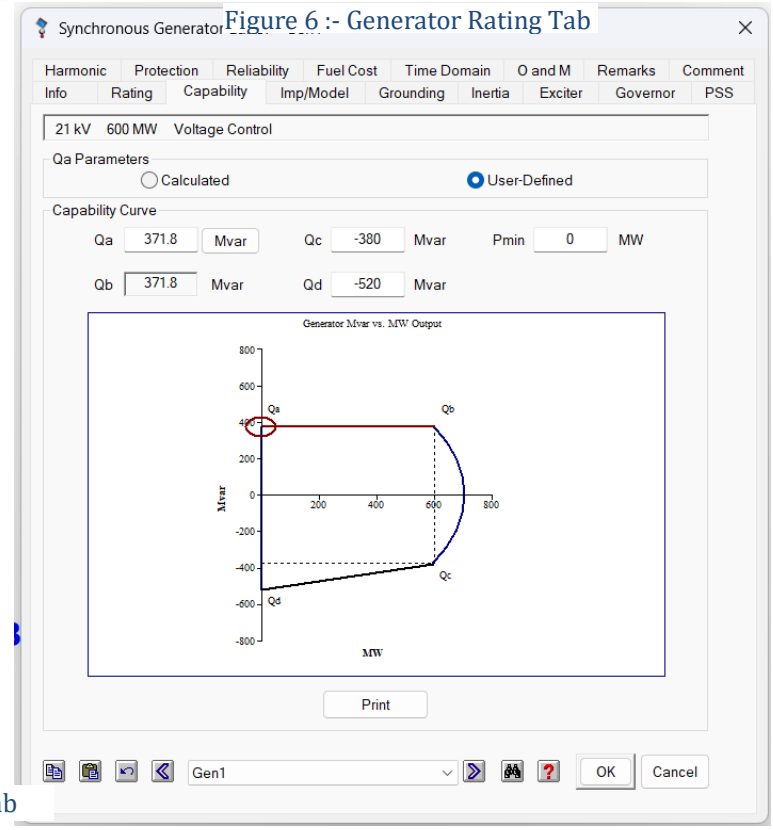


Figure 6 :- Generator Rating Tab



FigFigure 8 :- Generator Capability Curve Tab

6.3 Transformer Modelling

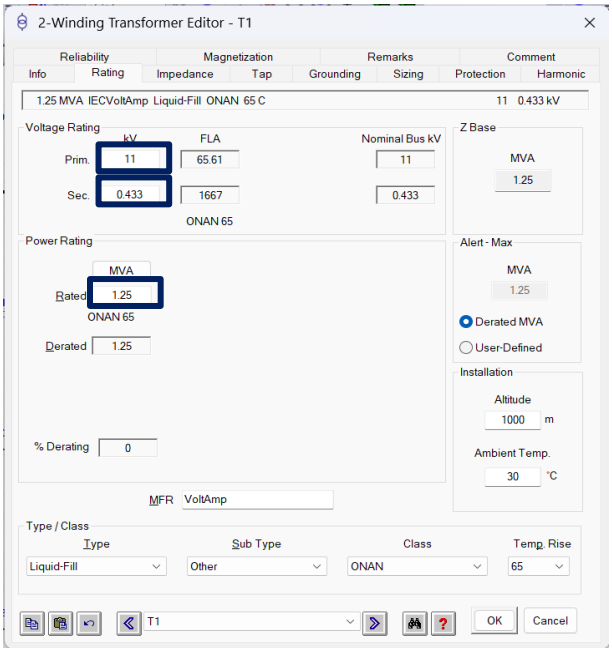


Figure 9 :- Transformer Rating Tab

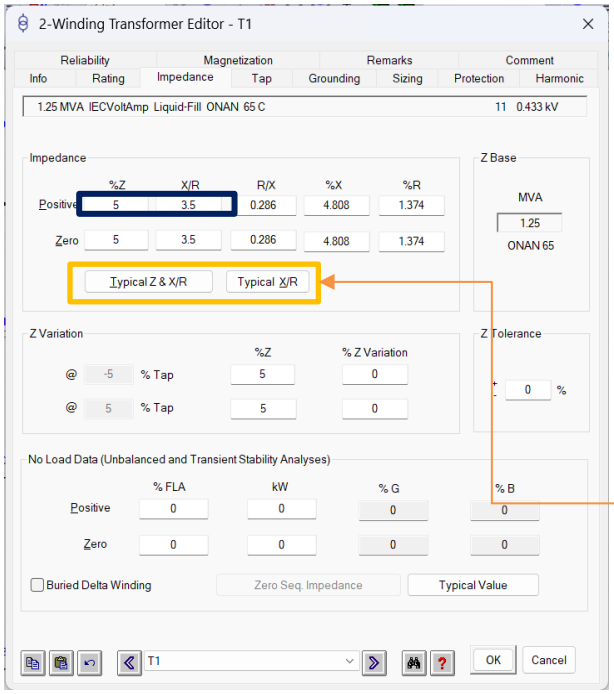


Figure 10 :- Transformer Impedance Tab

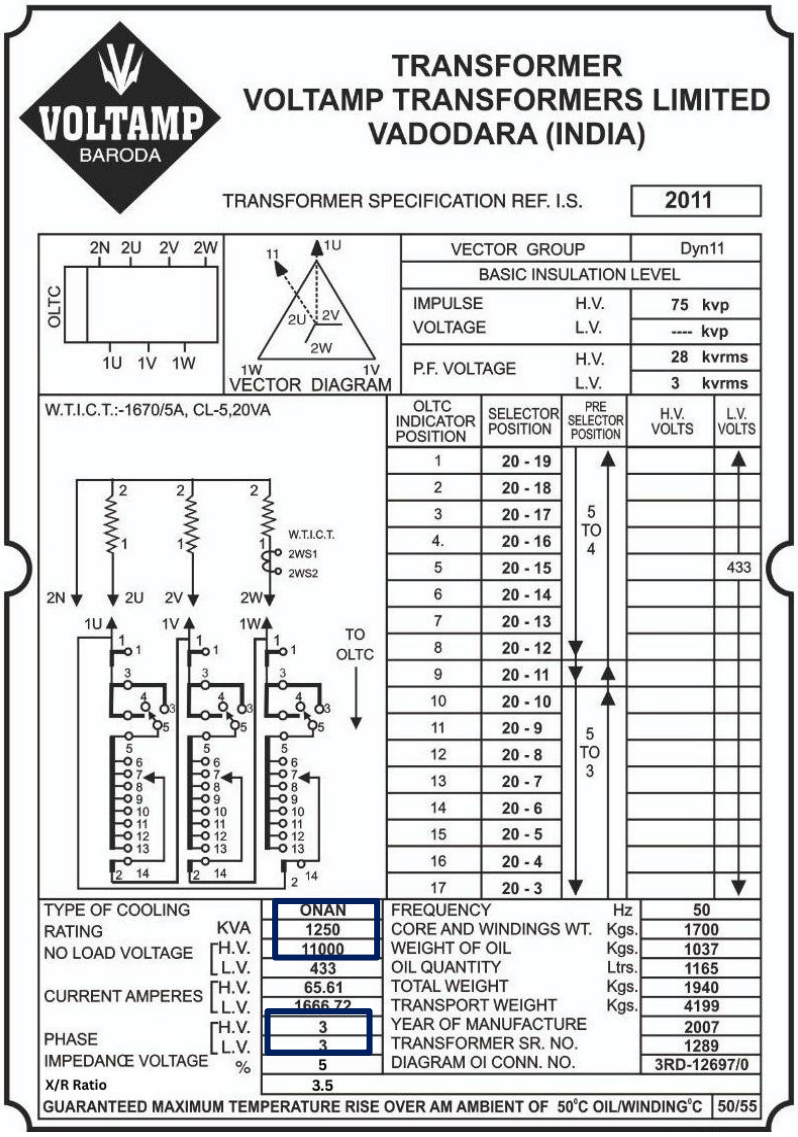


Figure 11 :- Transformer Name Plate

Industry-average values used when specific manufacturer data for a component is not yet available.

6.4 Circuit Breaker Modelling

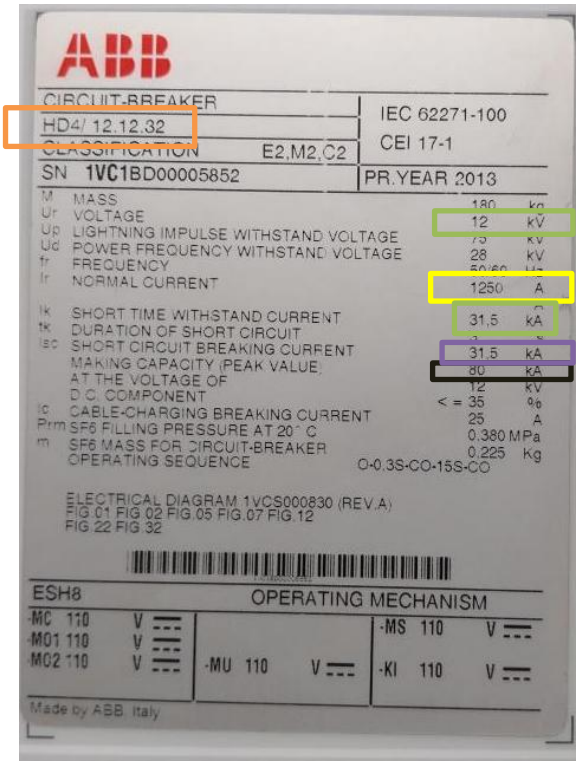


Figure 12 :- Circuit Breaker Nameplate

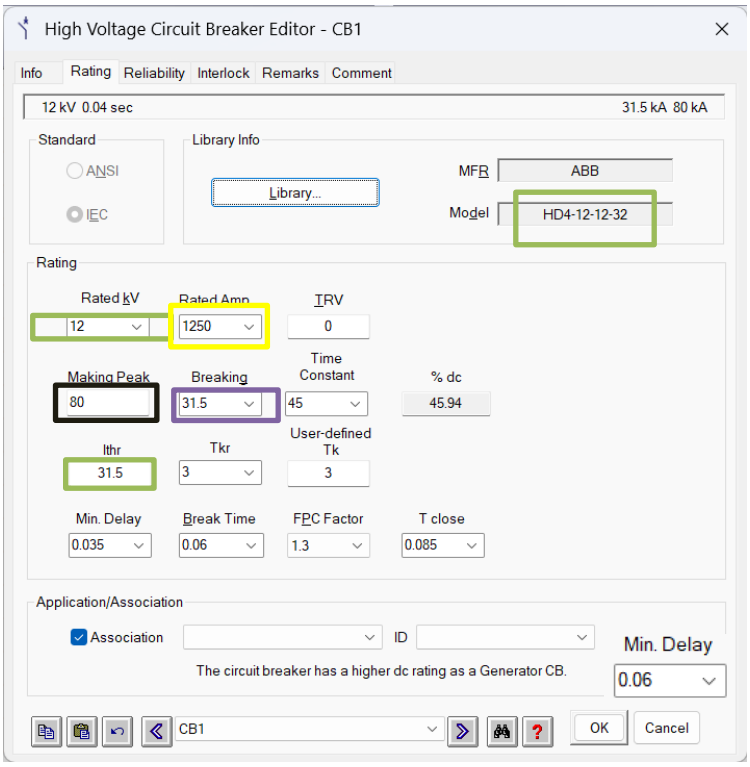


Figure 13 :- Circuit Breaker Ratings Tab

6.5 Current Transformer

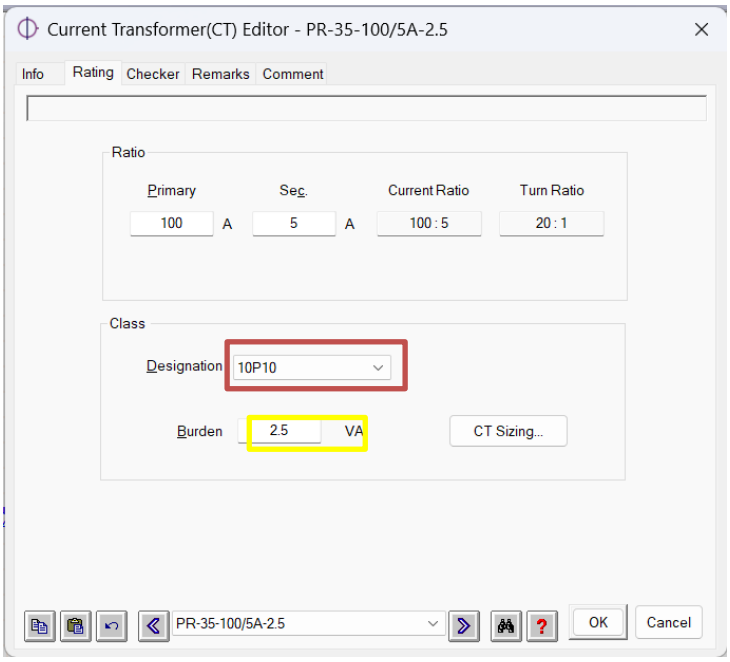
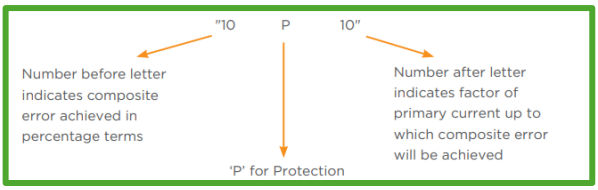


Figure 14 :- CT Ratings



CT burden is the total resistance (impedance) of the external circuit connected to the secondary terminals of a current transformer, which determines how much power the CT must supply to drive the secondary current.

6.6 Induction Motor Modelling

SIEMENS	Data Sheet : Induction Motor Champion Series Motor Type : 1SE0 319-4NA80			
Number of Phases		3		
Rated Output	kW	200		
Rated Voltage	V	415		
Rated Current	A	336	Reference Standard	IS:12615
Rated Speed	rev/min	1485	Enclosure IS:4691	IP55
Rated Frequency	Hz	50	Cooling IS:6362	IC411
Rated Torque	131 kgf.m / 1286 Nm		Duty	S1
Efficiency :	100% load %	95.1	Terminal Box Type	1XB7 622
	75% load %	95.1	Terminal Size	M16
	50% load %	95.1	No. of Terminals	6
Power Factor :	100% load %	0.87	Max. Cable Section mm²	2R 3½C 300 mm²
	75% load %	0.86		
	50% load %	0.81	Drive End Bearing Data	6319 C3
Insulation Class		F	Non-Drive End	6319 C3
Temperature rise (Res.method) K		80	Lubrication	Re-Greasable
			Life L10 hrs. #	50,000
Locked rotor withstand time	Hot (s)	15	Lifting eyebolt IS:4190	M24
	Cold (s)	25	Motor weight (approx.) kg	1220
Stator connection		DELTA	Rotor GD² (4x MI) kgf.m²	16.8
Rotor Class		KL16	Rotor Weight kg	346
			Construction IS:2253	IMB3
Ratio of starting current to the rated current	100% voltage	6.5	Frame designation	315L
	80% voltage	4.8		
			Ambient Temperature °C	50
Ratio of starting torque to the rated torque	100% voltage	2.8	Voltage Variation %	± 10
	80% voltage	1.6	Frequency variation %	± 5
			Combined Voltage & Frequency variation %	10
Ratio of Breakdown torque to the rated torque	100% voltage	2.6	# For directly coupled drives	
	80% voltage	1.5	Data subject to tolerance as per IS:12615.	

Figure 15 :- Induction Motor Datasheet

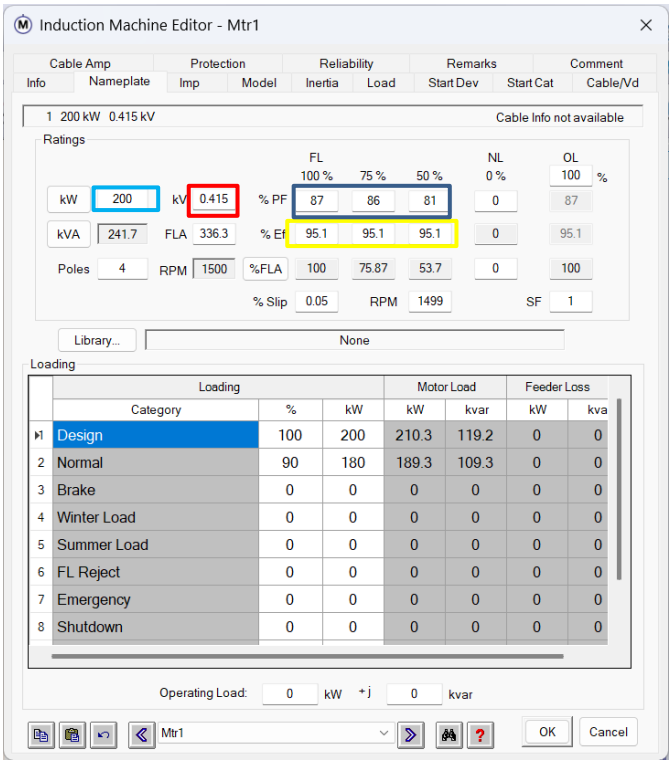


Figure 16 :- Induction Motor Nameplate Tab

6.7 Lumped Load Modelling

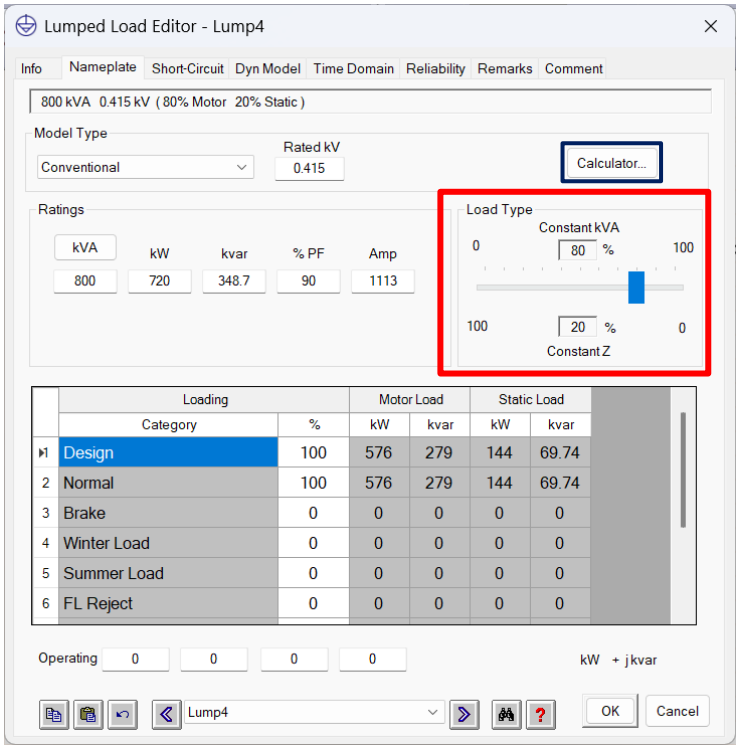


Figure 17 :- Lumped Load Nameplate

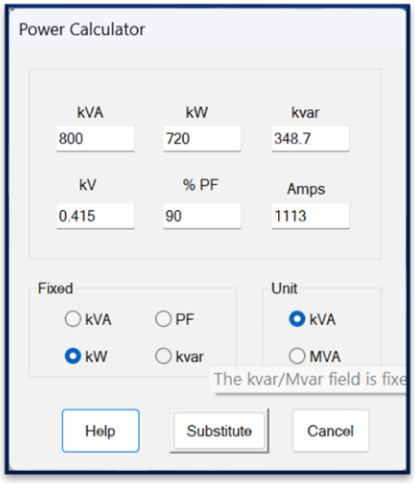


Figure 18 :- Lumped Load Calculator Tab

The **Load Type** indicates whether a load is dynamic (Motor) or passive (Static), determining how it reacts to voltage changes and contributes to faults. The **80% and 20%** split specifies that the majority of the 800 kVA is comprised of motors, while the remainder consists of resistive elements

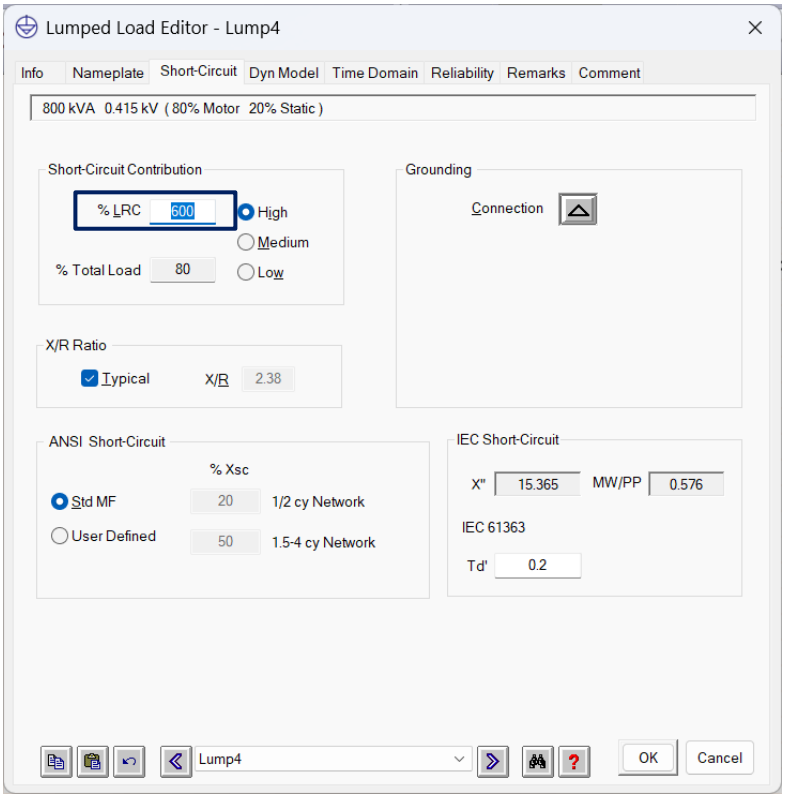


Figure 19 :- Lumped Load SC Tab

LRC, or Locked Rotor Current, usually indicates the motor's starting current, often at 600% or six times the rated current. This multiplier helps the software estimate the sub-transient reactance, which influences the initial fault current.

Typical Values :-

1. DOL Starter ≈6 × Nominal Current
2. Soft Starter ≈3 × Nominal Current
3. VFD ≈1 × Nominal Current (Negligible Contribution)

If a lumped load has 120 Amps total at 80% motor loading, the motor current is 96 Amps.

- At 600% LRC, the actual fault contribution is 6.0×96 A which is 576 Amps.

6.8 Cable Modelling

TECHNICAL DETAILS FOR HAVELLS 3.8/6.6 KV THREE CORE, ALUMINIUM/COPPER CONDUCTOR. XLPE INSULATED. ARMOURED CABLES

Size (Cross Sectional Area)	Max. Conductor D.C.Resistance at 20 °C		Approx. Conductor A.C.Resistance at 90 °C		Reactance of Cable at 50 Hz (Approx.)	Capacitance of Cable (Approx.)	Normal Current Rating						Short Circuit Current Rating for 1 Second Duration	
							For Aluminium Conductor			For Copper Conductor				
	Aluminium	Copper	Aluminium	Copper			Ground	Duct	Air	Ground	Duct	Air	Aluminium	Copper
Sqmm	Ohm/Km	Ohm/Km	Ohm/Km	Ohm/Km	Ohm/Km	µF/Km	Amps	Amps	Amps	Amps	Amps	Amps	K.amps	K.amps
25	1.20	0.727	1.54	0.931	0.164	0.18	100	90	120	130	115	155	2.35	3.58
35	0.868	0.524	1.11	0.671	0.156	0.20	120	105	145	155	140	185	3.29	5.01
50	0.641	0.387	0.820	0.495	0.147	0.22	140	125	170	185	160	220	4.70	7.15
70	0.443	0.268	0.567	0.343	0.139	0.26	175	155	215	225	195	275	6.58	10.01
95	0.320	0.193	0.410	0.248	0.133	0.29	205	180	260	265	235	340	8.93	13.59
120	0.253	0.153	0.325	0.197	0.127	0.32	235	205	305	300	265	390	11.28	17.16
150	0.206	0.124	0.265	0.159	0.124	0.35	260	230	345	335	295	440	14.10	21.45
185	0.164	0.0991	0.211	0.127	0.120	0.38	295	260	395	380	330	510	17.39	26.46
240	0.125	0.0754	0.162	0.0976	0.117	0.43	340	300	470	435	380	600	22.56	34.32
300	0.100	0.0601	0.130	0.0778	0.113	0.46	385	335	540	490	425	680	28.20	42.90
400	0.0778	0.0470	0.1023	0.0618	0.110	0.53	440	380	630	550	480	790	37.60	57.20
500	0.0605	0.0366	0.0808	0.0489	0.107	0.59	495	430	730	610	530	910	47.00	71.50
630	0.0469	0.0283	0.0648	0.0391	0.104	0.66	560	480	840	680	580	1030	59.22	90.10
800	0.0367	0.0221	0.0530	0.0319	0.100	0.74	620	530	960	740	630	1140	75.20	114.40
1000	0.0291	0.0176	0.0444	0.0268	0.098	0.82	680	580	1070	790	670	1250	94.00	143.00

Figure 20 :- Snapshot of 6.6kV Cable from Havells Catalogue

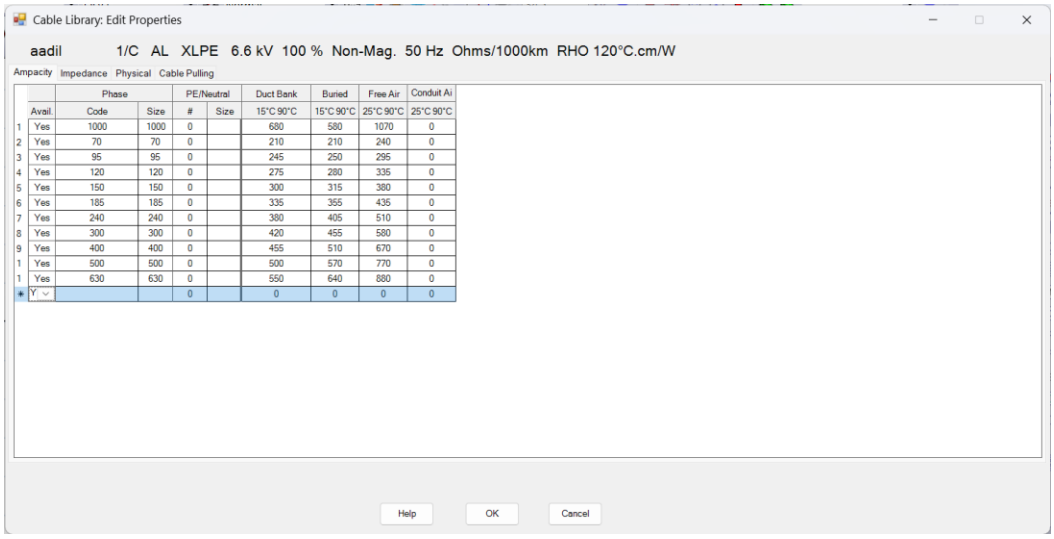
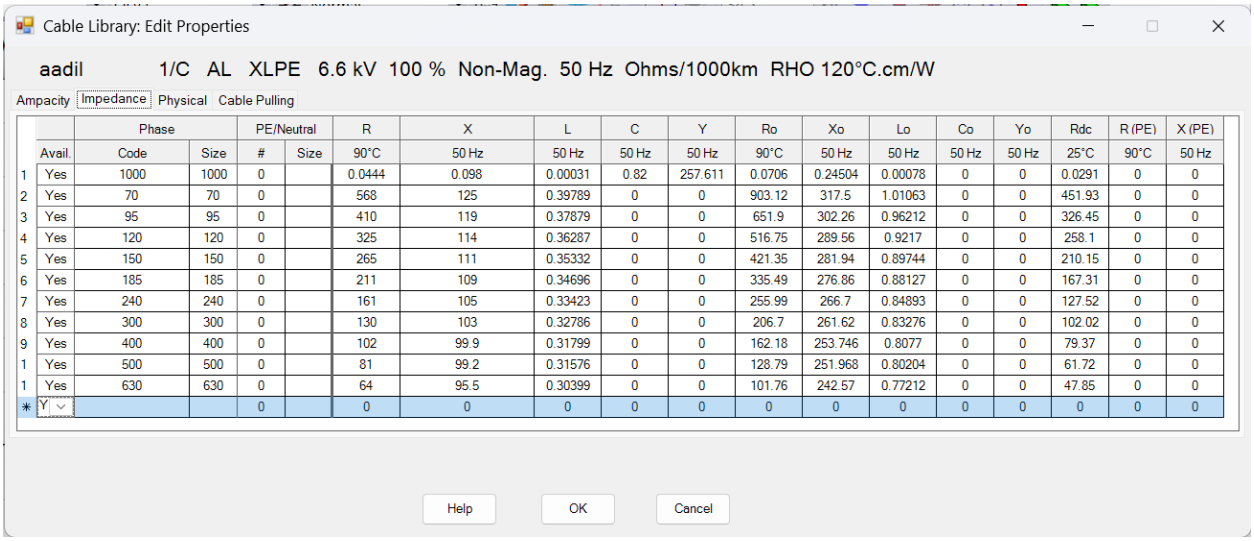


Figure 21 A :- Inserting Values from Catalogue in ETAP Cable Modelling



	Avail.	Phase Code	Phase Size	PE/Neutral #	PE/Neutral Size	R 90°C	X 50 Hz	L 50 Hz	C 50 Hz	Y 50 Hz	Ro 90°C	Xo 50 Hz	Lo 50 Hz	Co 50 Hz	Yo 50 Hz	Rdc 25°C	R (PE) 90°C	X (PE) 50 Hz
1	Yes	1000	1000	0		0.0444	0.098	0.00031	0.82	257.611	0.0706	0.24504	0.00078	0	0	0.0291	0	0
2	Yes	70	70	0		568	125	0.39789	0	0	903.12	317.5	1.01063	0	0	451.93	0	0
3	Yes	95	95	0		410	119	0.37879	0	0	651.9	302.26	0.96212	0	0	326.45	0	0
4	Yes	120	120	0		325	114	0.36287	0	0	516.75	289.56	0.9217	0	0	258.1	0	0
5	Yes	150	150	0		265	111	0.35332	0	0	421.35	281.94	0.89744	0	0	210.15	0	0
6	Yes	185	185	0		211	109	0.34696	0	0	335.49	276.86	0.88127	0	0	167.31	0	0
7	Yes	240	240	0		161	105	0.33423	0	0	255.99	266.7	0.84893	0	0	127.52	0	0
8	Yes	300	300	0		130	103	0.32786	0	0	206.7	261.62	0.83276	0	0	102.02	0	0
9	Yes	400	400	0		102	99.9	0.31799	0	0	162.18	253.746	0.8077	0	0	79.37	0	0
1	Yes	500	500	0		81	99.2	0.31576	0	0	128.79	251.968	0.80204	0	0	61.72	0	0
1	Yes	630	630	0		64	95.5	0.30399	0	0	101.76	242.57	0.77212	0	0	47.85	0	0
*	Y			0		0	0	0	0	0	0	0	0	0	0	0	0	0

Figure 21 B :- Inserting Values from Catalogue in ETAP Cable Modelling

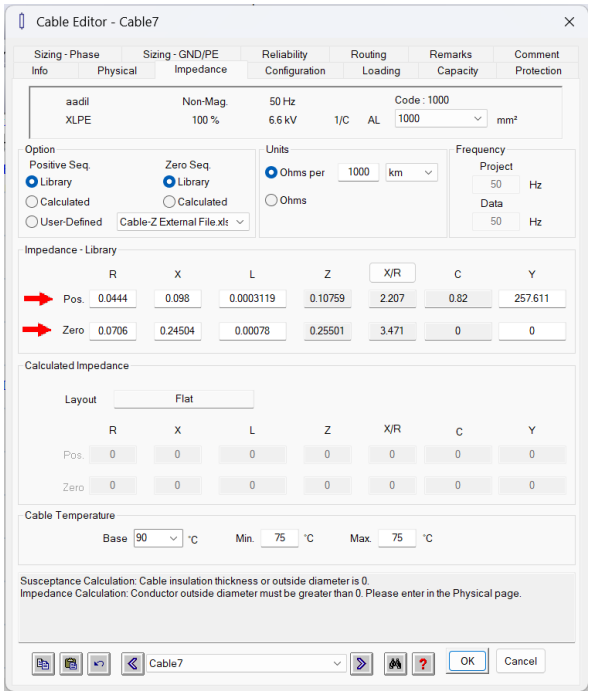


Figure 22 :- Verifying the values in ETAP Tab

Check the following to verify the cable modelling

$R_0/R = 1.59$

$X_0/X = 2.5$

Chapter - 07 Load Flow Study

7.1 Introduction

A Load Flow Study checks how the electrical system in a mineral processing plant works when its steady. The main goal is to see active power (P), reactive power (Q), apparent power (S), current, and voltage move across the network. This ensures that big machines like grinding mills and crushers work properly.

7.2 Purpose of Load Flow Study

For this mineral processing facility, the study does the following:

a) Determine power and current flows in the 220/11.5 kV and 11/0.433 kV transformers.
b) Make voltage levels at the 11 kV and 415 V Motor Control Centers (MCCs) are stable when production is high.
c) Find transformer tap-settings needed to fix voltage drops caused by long cable runs to remote mine-site loads.

Check current ratings against the total plant demand of about 25-30 MW.

7.3 How to Do a Load Flow Study

To do a load flow study you need to follow these steps:

1. Collect all the information like the Single Line Diagram, data sheets for electrical equipment and Factory Acceptance Test reports if you have them.
2. Make a model of the system using a software like ETAP using the information you collected.
3. Think about situations that could happen like changes in load, voltage fluctuations, equipment failures and what the client wants.
4. Run simulations for each situation to see how the system behaves under conditions.
5. Put together all the results, including voltage levels, loading, losses and any problems or areas that need improvement.
6. Make a report and present your findings along with suggestions for design, settings or operational improvements.

7.4 What You Get from a Load Flow Study

A load flow study gives you information about how an electrical system works. You use this information to check if the system is stable find problems and make decisions about design, operation and optimization. The main things you get from a load flow study are:

1. Voltage Profile:

- The voltage level and phase angle at each point in the system.
- Making sure all voltage levels are within the allowed limits usually plus or minus 5% of the rated voltage.

2. Power Flow Analysis:

- How much power is flowing through all the lines and transformers.
- Making sure the power flow is balanced and no part of the system is overloaded.

3. Current Distribution:

- How much current is flowing through each line, transformer and piece of equipment.
- Finding out if any equipment is overloaded or underused.

4. Losses in the System:

- How much power is lost due to resistance in conductors and transformers.
- How much reactive power is lost due to system inductance and capacitance.

5. Power Factor Evaluation:

- The power factor at each point in the system making sure it is within the allowed range more than 0.95 for system efficiency.

6. Equipment Rating Confirmation:

- Checking if transformers, switchgear, cables and other equipment are the size and rating for their intended use.

7. System Performance Under Contingencies:

- How the system behaves when something fails or is out of service like a transformer or line failure or a change in grid voltage.
- Making sure the system stays stable and works even if something goes wrong.

8. Optimization Recommendations:

- Suggestions for improving efficiency and stability like using voltage regulation devices, power factor correction equipment and transformer tap settings.

9. Future Expansion Feasibility:

- Checking if the system can handle growth without needing significant changes or overloading existing parts.

In short, a load flow study helps make sure the electrical system works efficiently within limits and can handle current and future demands while minimizing losses and optimizing equipment performance.

7.5 What is IEEE 3002.2 ?

IEEE 3002.2 is a standard for doing load flow studies in commercial power systems. It provides guidelines for system performance, efficiency, reliability and protection.

IEEE 3002.2 gives guidelines for making sure industrial and commercial electrical systems work efficiently and reliably. It highlights practices for load flow analysis, voltage regulation, transformer and cable sizing and protection coordination. By following this standard engineer can design systems that're stable, safe and optimized for energy efficiency.

7.5.1 Key Aspects of IEEE 3002.2

1. **Load Flow Analysis:**

- A key goal of the load flow study is to ensure that voltage levels at all buses (especially distribution buses like 11 kV) stay within acceptable limits under normal and peak load conditions.
- Voltage Range: The standard suggests that voltage at any bus should ideally remain between ±5% of its nominal value under normal operating conditions. For example, for a 400V bus, the voltage should stay between 380V and 420V.
- The study also considers reactive power and real power flows to identify voltage drops and overloading.

2. **System Efficiency & Loss Minimization:**

- IEEE 3002.2 emphasizes the need to minimize losses in cables and transformers. The standard recommends calculating I^2R losses in distribution systems to ensure efficient operation.
- **Transformer Loading**: It is advised that transformers should operate at no more than 80%-85% of their rated capacity under normal conditions to ensure thermal limits are not exceeded.

3. **Voltage Regulation:**

- Proper voltage regulation is crucial. The standard encourages maintaining voltage stability within the specified limits for continuous operation. In systems with distribution transformers, voltage regulation should ideally be within ±5% of the nominal value to prevent operational disruptions.

4. **Protection Coordination:**

- The standard provides guidelines for protection devices such as circuit breakers and relays to ensure selective tripping during faults. Protection devices should be properly coordinated to ensure instantaneous protection in case of short circuits or overcurrent conditions.
- **Time-Current Characteristics**: Protection devices must meet specific time-current characteristics, typically defined for devices like fuses and relays. The time delay for trip settings should be in the range of 0.1 to 1 second for short circuits and up to several seconds for overload protection.

5. **System Data & Accuracy:**

- Accurate system data, including load magnitude, impedance, generator ratings, and line characteristics, is necessary for reliable load-flow studies. Proper input data ensures the analysis results will accurately reflect real-world performance.

7.5.2 Design and Sizing Recommendations

1. **Transformer Sizing:**
- For reliable operation, transformers should be sized to handle the expected peak load, plus an additional 20% margin to account for future growth or transient conditions.
2. **Cable Sizing:**
- Cables should be selected to handle maximum expected current while keeping voltage drop under control. A voltage drop of no more than 2-3% from the source to the load is typically recommended.
3. **Power Generation:**
- If generators are part of the system, their generation capacity should be sized to handle peak loads with a margin of around 10-15% over the expected maximum load to ensure stable operation during load fluctuations.
4. **Busbar Sizing**:
- Busbars should handle full load current with a **20-25% margin** for future growth and transient conditions, and maintain a **<50°C** temperature rise to prevent overheating.
5. **Protection Devices**:
- Protection devices must be sized for **overload** and **short-circuit** conditions, with **short-circuit protection** tripping in **0.1-1 second** and **overload protection** delayed for system coordination to ensure selective tripping.
6. **Power Factor Correction**:
- Maintain a **0.95** power factor or higher for efficiency. Use **capacitor banks** or **synchronous condensers** to correct power factor and reduce load on transformers and generators.

7. **Energy Storage Systems**:
   - Energy storage (e.g., batteries) supports **peak shaving** and smooths fluctuations in renewable energy, improving r**eliability** and **efficiency.**

7.5.3 Specific Ranges Mentioned in IEC 3002.2

- **Voltage Tolerance**: Voltage at any bus: **±5%** of nominal voltage under normal conditions.
- **Transformer Loading**: Transformers should ideally not exceed **85%** of their rated capacity under normal operation.
- **Cable Voltage Drop**: Voltage drop should typically remain **≤ 2-3%** to ensure efficient power delivery.
- **Protection Device Time-Current Characteristics**: Short-circuit protection devices: **0.1 to 1 second** trip time.
- **Overload protection:** up to several seconds, depending on the protection scheme and system characteristics.

7.6 System Description

The mineral processing plant gets power at 132 kV from the utility grid. This high-voltage supply is reduced to 11 kV via two 25/31.5 MVA ONAN/ONAF transformers (TR-1 and TR-2).

The distribution network is divided into key zones:

- **Primary Distribution :** The 11 kV Main Bus is rated for 3000A. is split into two sections connected by a 3000A Bus Coupler.
- **Captive Power Integration :** The system includes a WHRS (Waste Heat Recovery System) Generator that provides 12 MW at 11 kV. This lets the plant work with the grid for energy efficiency.
- **Secondary Distribution :** Power is reduced to 415V for loads, through various transformers.
- **Grounding Philosophy :** The 11 kV system neutral is grounded through a Neutral Grounding Resistor (NGR). This limits fault to 100A to protect sensitive motor windings.

Chapter-08 Short Circuit Study

8.1 Introduction

Short circuit analysis is a crucial part of power system studies. It determines the size of fault currents under different conditions, such as three-phase, line-to-line, and single line-to-ground faults. In this project, we used ETAP software for a detailed short circuit study of the cement plant's electrical distribution network. This analysis helps evaluate the fault levels at different buses and makes sure that all equipment, including circuit breakers, transformers, and cables, are rated correctly to safely handle and interrupt fault currents.

After the fault analysis, we conducted a Device Duty (DD) Check. This check verifies that the interrupting capacities of protective devices remain within limits during maximum fault conditions. It ensures reliable and safe operations during abnormal events.

We also performed a Relay Coordination Study. This study ensures proper grading between protective devices like relays, breakers, and fuses. The goal is to isolate faults as close to where they occur as possible. This keeps the system stable and minimizes disruption to healthy parts of the network. It is essential for preventing cascading failures and improving the overall performance of the protection system.

In ETAP, short circuit studies are performed under two main scenarios:

Device Duty (DD) for evaluating maximum fault current and **Relay Coordination Duty (RC)** for minimum fault current evaluation. The study uses different tolerance applications based on the chosen options.

Device Duty (DD) – Max

Negative tolerance is applied to:

Generator & motor sub-transient reactance (Xd")

Transformer, reactor, and overload heater impedances

Cable and line lengths (if tolerance on length is selected)

Resistance is adjusted based on the minimum operating temperature to reduce resistance and simulate the worst-case scenario for maximum fault current.

Relay Coordination Duty(RC) – Min

Positive tolerance is applied to:

Generator & motor sub-transient reactance (Xd")

Transformer, reactor, and overload heater impedances

Cable and line lengths

Resistance is adjusted based on the maximum operating temperature, increasing resistance and leading to lower fault current for coordination analysis.

8.2 Purpose of Short Circuit Study

The Short Circuit Study aims to evaluate the system's response to fault conditions and ensure protection is safe, reliable, and coordinated. Key objectives include:

- Determine maximum fault currents at various buses to assess worst-case fault scenarios.
- Verify that protective devices like circuit breakers and fuses meet interrupting ratings against fault levels.
- Ensure proper coordination of protection settings for selective fault isolation without affecting healthy system sections.
- Identify under-rated or overstressed equipment that may need upgrades for safety during fault conditions.
- Evaluate Device Duty (DD) by comparing maximum fault currents with equipment withstand ratings for mechanical and thermal stability.
- Perform Relay Coordination (RC) by analyzing minimum fault currents to validate the sensitivity and proper operation of protection relays during low-magnitude faults.

8.3 Types of Faults Analyzed

In power systems, faults disrupt the normal current flow, potentially damaging equipment and destabilizing the system. Faults are mainly classified into open circuit faults and short circuit faults.

1. Open Circuit Faults

Open circuit faults happen when there is a break in the conducting path, interrupting the current flow. Also known as series faults, they can be further categorized into:

- Single Conductor Open Fault: One phase conductor is broken.
- Two Conductors Open Fault: Two phase conductors are broken.
- Three Conductors Open Fault: All three phase conductors are broken.

While less severe than short circuit faults, open circuit faults can still impact system reliability and are important for analysis.

2. Short Circuit Faults

Short circuit faults occur when conductors of different phases come into contact with each other or the ground, causing excessive current flow. These faults are divided into:

a. Symmetrical Faults

These involve all three phases equally and are balanced in nature.

- Three-Phase Fault (L-L-L): All three phases are shorted together. This is the most severe fault type, causing the highest fault currents. Although rare (about 2–5% of all faults), they are crucial for determining ratings of protective devices like circuit breakers.
- Three-Phase to Ground Fault (L-L-L-G): All three phases are shorted to each other and to ground. This fault is also severe and is used for sizing equipment and protection studies.

b. Unsymmetrical Faults

These involve one or two phases, leading to unbalanced system conditions.

- Single Line-to-Ground Fault (L-G): One phase conductor contacts the ground. This is the most common fault type, making up about 70–80% of all faults. It has a significant impact on system stability and protection schemes.
- Line-to-Line Fault (L-L): Two phase conductors contact each other. This fault type represents about 10–15% of faults and can be caused by external factors like high winds that cause conductors to touch.
- Double Line-to-Ground Fault (L-L-G): Two phase conductors come into contact with each other and the ground. This type accounts for about 10% of faults and combines the effects of both L-L and L-G faults.

8.4 Steps to Perform Short Circuit Study in ETAP

1. Model the system with all components (generators, transformers, loads, etc.).
2. Select the Short Circuit Study Case (IEC/ANSI, Max/Min fault levels).
3. Set parameters like tolerances, c-factor, and X/R ratio method.
4. Choose fault locations (buses, branches).
5. Run the SC study and review bus bracing, device ratings, and relay behavior.
6. Take action if any component is overstressed or fails coordination.

8.5 IEC 60909 Overview

8.5.1 Introduction

The IEC 60909 standard provides guidelines for calculating short-circuit currents in three-phase AC systems. It uses the impedance method to determine fault levels under worst-case conditions, ensuring safe and reliable equipment selection and system design. This standard considers both maximum and minimum fault conditions, taking into account variations in system setup, source strength, and equipment tolerances.

8.5.2 Purpose

The main purpose of conducting a short-circuit study based on IEC 60909 is to:

- Determine the maximum short-circuit current for equipment rating and mechanical strength.
- Determine the minimum short-circuit current for relay coordination and protection sensitivity.
- Ensure all power system components, like switchgear, transformers, cables, and protection devices, are rated correctly to withstand and interrupt fault conditions.
- Provide a basis for coordinating protection systems and ensuring safe network operation.
- Support engineering decisions for system expansion, modification, or fault mitigation.

8.5.3 Key Objectives

- Perform short-circuit current calculations for symmetrical and asymmetrical faults using IEC 60909 methodology.
- Identify buses or sections where fault current levels exceed equipment interrupting capacity.
- Help set protective relays and verify device coordination under minimum fault conditions.

❖ Recommend mitigation techniques such as:

1. Current limiting reactors (CLR)
2. High impedance transformers
3. Network segmentation (bus coupler operation)
4. Cable impedance management
5. Ensure system design meets compliance with IEC 60909 for operational safety and reliability.

Steps For Device Duty

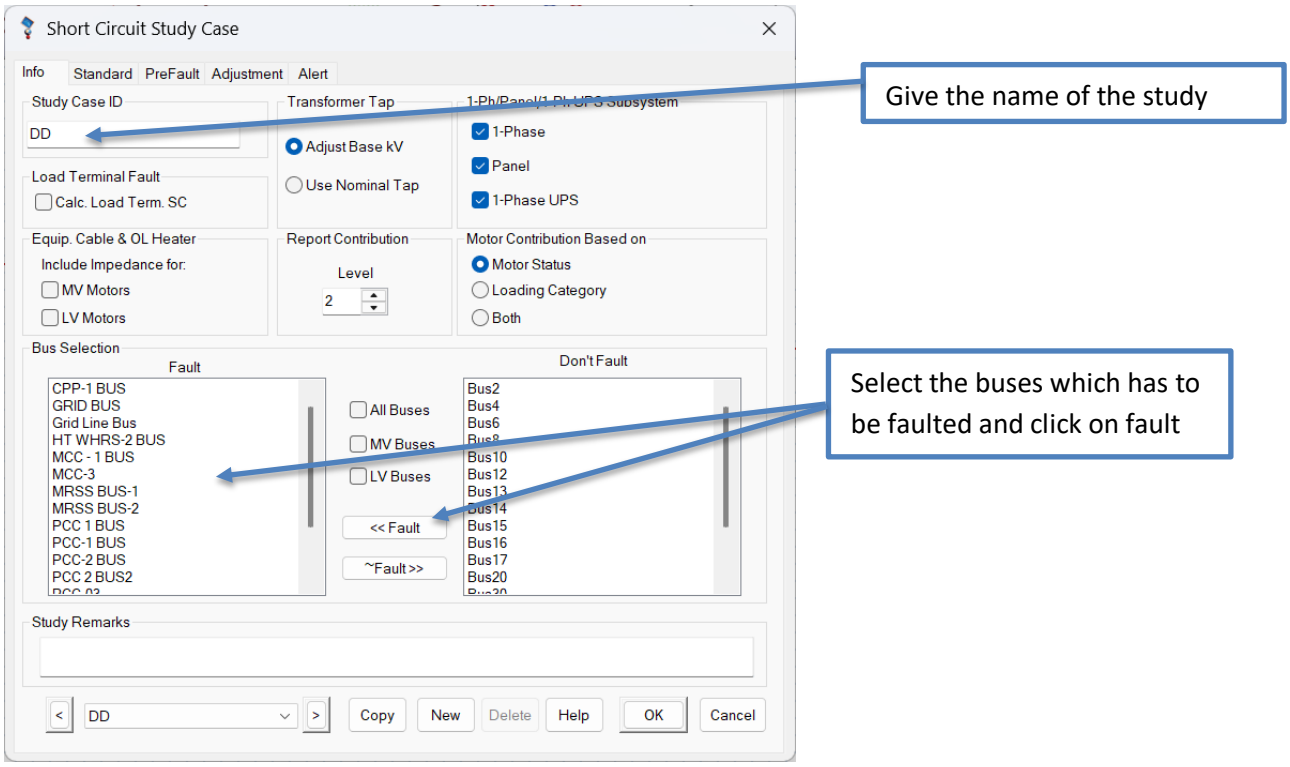


Figure 23 :- Device Duty Case

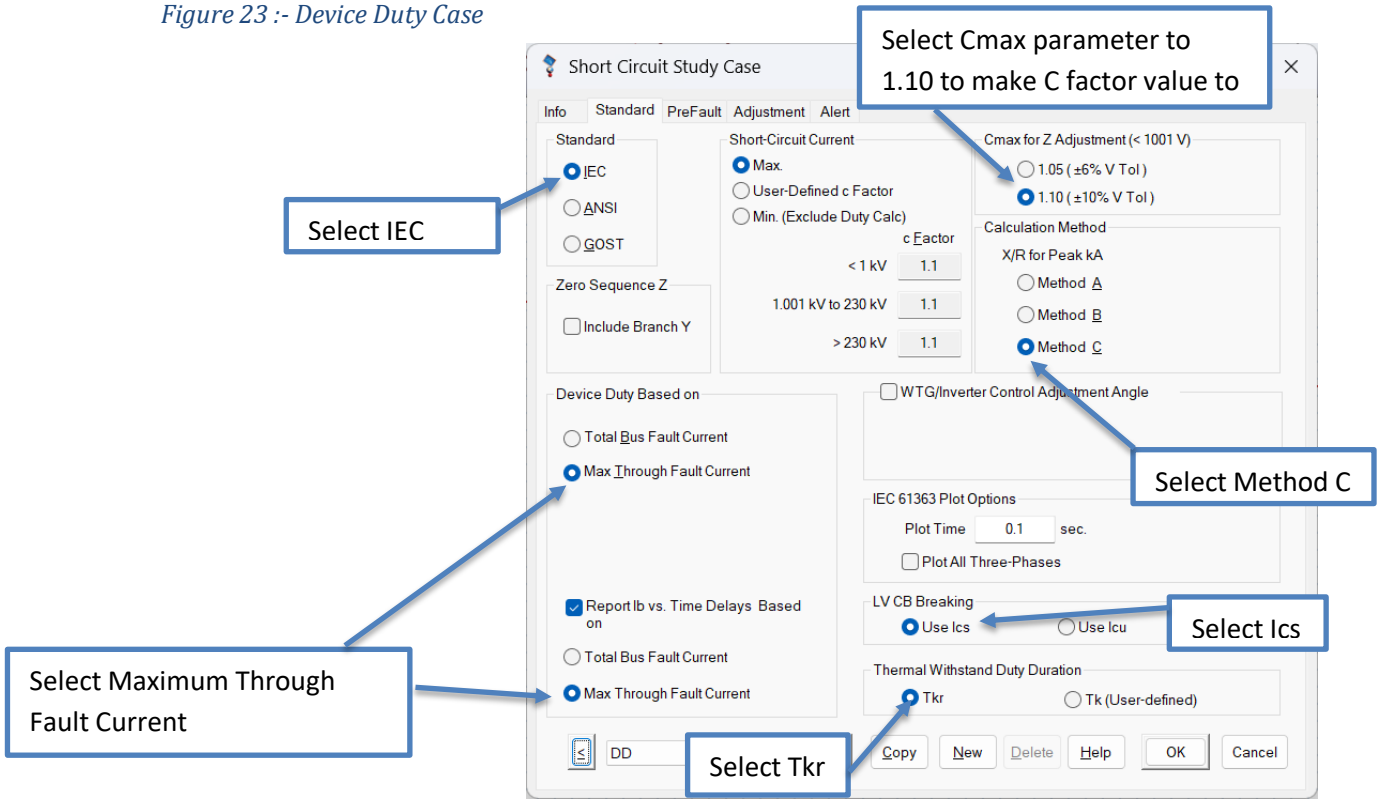


Figure 24 :- Device Duty Settings

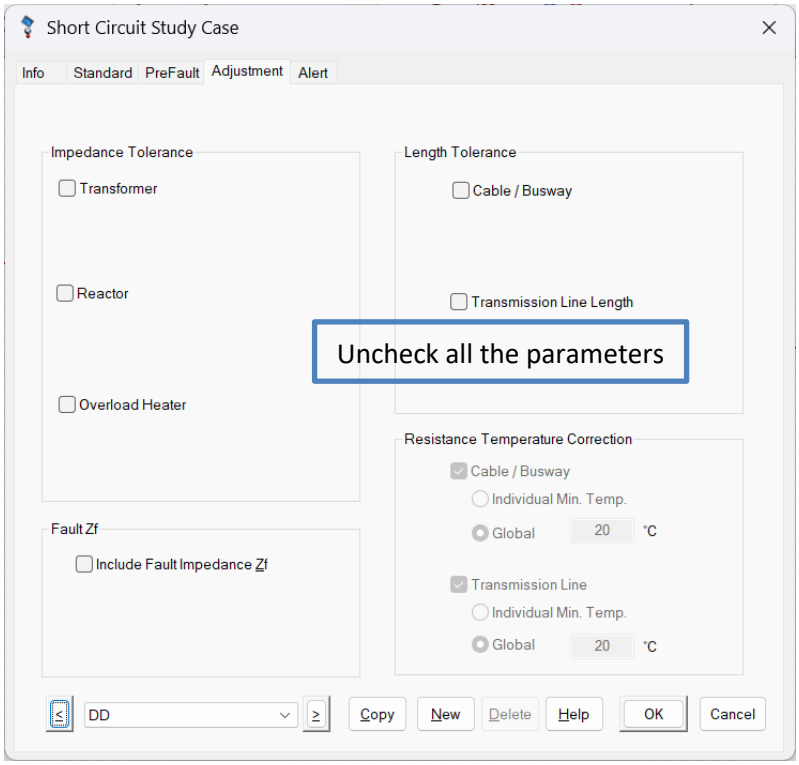
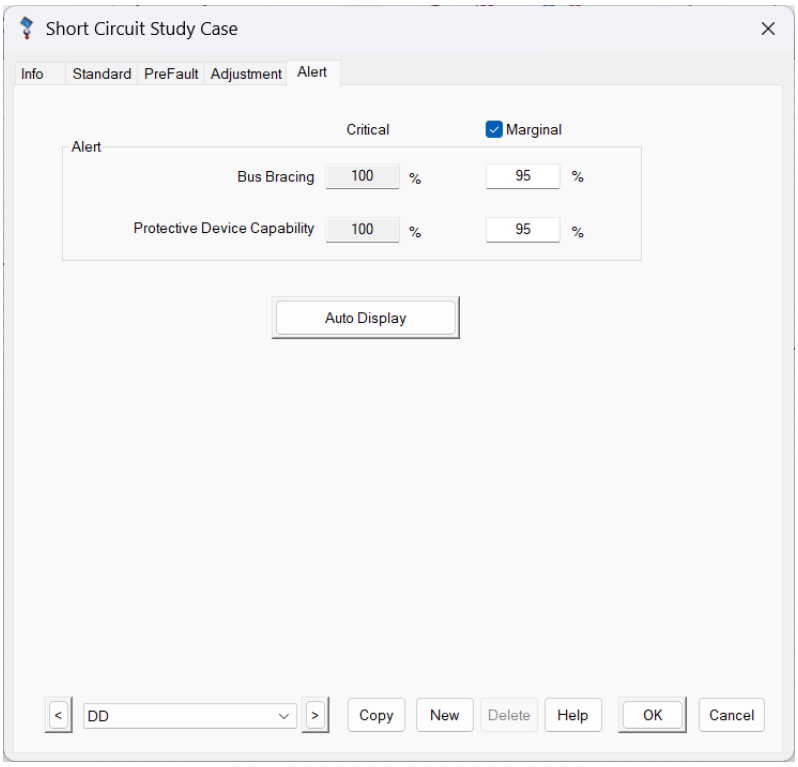


Figure 25 :- Adjustment Setting in Device Duty

- In a Green Field Project, actual site data is not available during the design stage. Therefore, tolerances are applied to estimated parameters such as impedance and cable length to account for possible variations during implementation
- Hence, in ETAP, it is necessary to select impedance tolerance and length tolerance under project settings to ensure realistic and conservative results in system studies.
- In a Brown Field Project, the electrical system already exists, and actual site data is available for all components such as cables, transformers, and loads.
- Therefore, there is no need to apply impedance or length tolerances, as precise values are used for short circuit, load flow, and protection coordination studies. This allows for more accurate and optimized analysis results.



It allows you to set alert thresholds for Bus Bracing and Protective Device Capability, defining when ETAP should flag a Critical (100%) or Marginal (95%) overload condition during fault analysis.

Figure 26 :- Alerts Settings in Device

Steps For Relay Co-Ordination

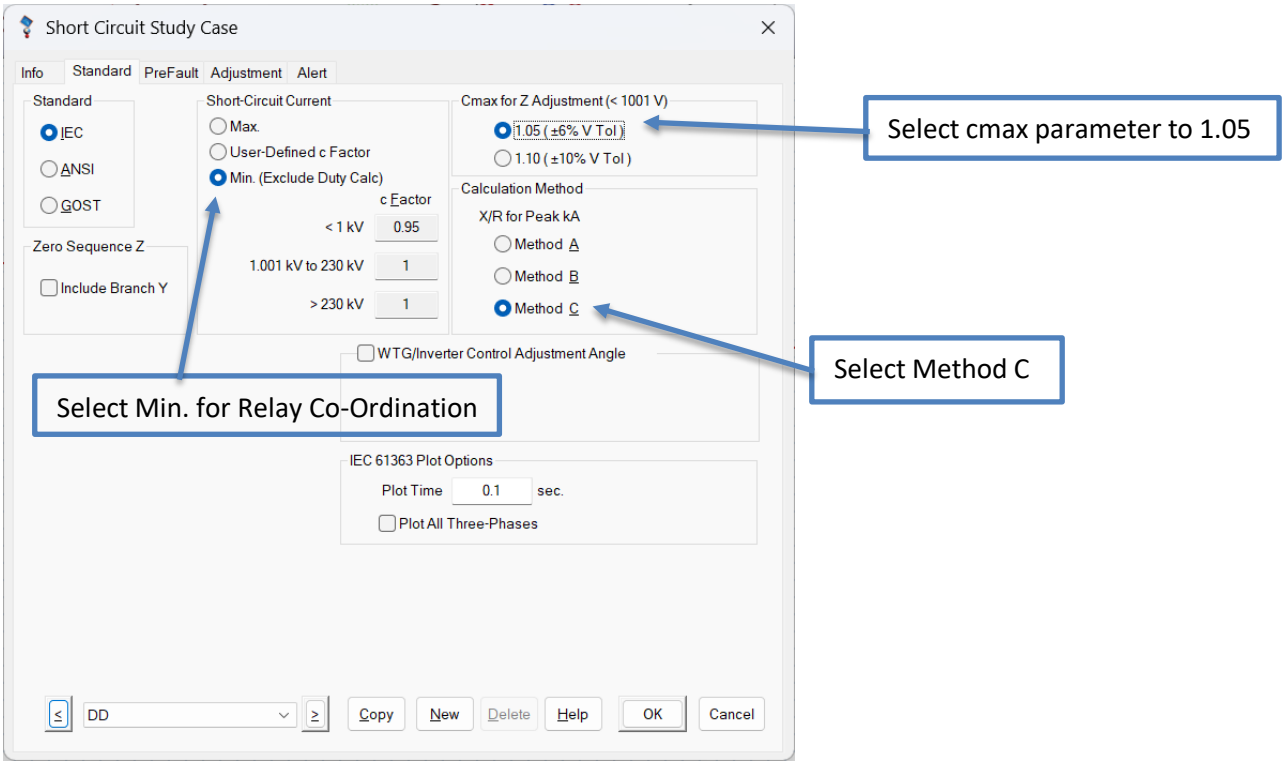


Figure 27 :- Standard Settings in Relay Co-Ordination

Chapter -09 Relay Co-Ordination Calculation

9.1 Introduction

Relay coordination plays a vital role in the protection of power systems by ensuring that only the faulty part of the network is isolated, which reduces disruption to the remaining system. Effective coordination among protective devices such as overcurrent relays, fuses, and circuit breakers guarantee that the nearest upstream device activates first during a fault, thereby preserving system stability and safety. In industrial power systems, particularly those with various voltage levels and sources like grid imports and captive generation (e.g., WHRS), it is crucial to achieve successful relay coordination to:

- Prevent unnecessary shutdowns
- Safeguard critical equipment
- Avoid cascading failures
- Ensure the safety of personnel

The coordination study includes evaluating the time-current characteristics (TCC) of protective devices in various fault scenarios to confirm grading margins, pickup settings, and delay timings.

This analysis, conducted with ETAP software, assesses the lowest fault currents (according to IEC 60909) to guarantee sensitivity and incorporates device information, relay characteristics, and system layout to ensure dependable and coordinated tripping. The Coordination Time Interval (CTI) indicates the minimum time gap necessary between the actions of primary and backup protective devices to maintain appropriate system selectivity. The CTI is affected by the following elements:

Table 1 :- Relay Time Grading

Component	Electromechanical Relay	Static Relay
Circuit Breaker Opening Time	0.08 s	0.08 s
Relay Overtravel	0.10 s	0.00 s
Relay Tolerance & Setting Errors	0.12 s	0.12 s
Total CTI Required	**0.30 s**	**0.20 s**

9.2 Time Grading Philosophy

Time grading serves as a fundamental method in power system protection. It ensures that only the affected section is swiftly disconnected while maintaining the operation of the remaining system. When executed properly, it offers dependable and economical protection coordination.

Reasons to Utilize Time Grading ?

1. Ensures selectivity by isolating only the faulted area.
2. Prevents unnecessary activation of upstream devices.
3. Minimizes system downtime and safeguards equipment.

Applications

1. Radial feeders in distribution systems.
2. Power networks in industrial facilities.
3. Protection for transformers and cables.
4. Networks operating at low and medium voltages.

Benefits of Time Grading

1. Straightforward and easy to apply.
2. Guarantees system coordination and stability.
3. Cost-efficient for radial configurations.
4. Lowers the risk of nuisance tripping.

Drawbacks

1. Not suitable for networks that are meshed or ring configurations.
2. Coordination becomes more complex as the system size increases.
3. Delays in time may result in increased fault energy affecting upstream devices.

9.3 What actually the Relay Co-Ordination is ?

Relay coordination is the systematic process of configuring protective devices to operate in a specific, timed sequence during a fault. The objective is simple: isolate the problem area as fast as possible while keeping the rest of the electrical grid energized. By precisely setting relays, the device closest to the fault trips first, while those further "upstream" (closer to the power source) act as backups. This strategy minimizes downtime, protects expensive equipment, and maintains overall system stability.

1. Core Coordination Techniques

To achieve selective protection, engineers use three primary grading methods:

- **Time Grading:** Relays are set with staggered operating times. The closer a relay is to the fault, the faster it trips. Upstream relays are intentionally slowed down to give downstream devices the first chance to clear the fault.
- **Current Grading:** Relays are set to trigger at different current thresholds. This is effective when fault current levels vary significantly across different parts of the system.
- **Zone Grading:** The system is divided into specific "zones" of protection. Relays are configured to monitor only their assigned area, ensuring that a fault in Zone A doesn't unnecessarily trip the breaker for Zone B.

2. Why Perform a Relay Coordination Study?

A formal study is essential for several technical and safety reasons:

- **Optimizing Settings:** To determine the best possible parameters for breakers, relays, and fuses.
- **Visual Validation:** To use **Time-Current Characteristic (TCC)** curves to ensure that upstream and downstream devices don't "overlap" or trip out of order.
- **System Changes:** To recalibrate protection when a plant expands, adds new motors, or changes its connection to the utility grid.
- **Safety & Arc Flash:** Coordination studies provide the "fault clearing time" data required to calculate incident energy for Arc Flash safety ratings.

3. Key Technical Terms

To understand how relays are tuned, you must know two critical variables:

Plug Setting Multiplier (PSM)

The PSM tells us how severe a fault is relative to the relay's setting. It is the ratio of the actual fault current to the relay’s "pickup" current.

- **High PSM:** Indicates a massive short circuit; the relay should trip quickly.

- **Low PSM:** Indicates a minor overload; the relay may wait longer or not trip at all.

$$\text{Plug Setting} = \frac{\text{Full Load Amperes}}{\text{CT Ratio}} \qquad \text{PSM} = \frac{\text{Fault Current}}{\text{CT Ratio} \times \text{Plug Setting}}$$

Time Multiplier Setting (TMS)

While the PSM is determined by the fault itself, the TMS is a setting we choose. It acts as a "speed dial" for the relay. A lower TMS makes the relay trip faster for any given current, while a higher TMS adds a delay to allow for coordination with other devices.

4. IDMT Curves: The "Inverse" Logic

Most modern protection relies on **IDMT (Inverse Definite Minimum Time)** characteristics. The logic is intuitive: the higher the current (the more dangerous the fault), the faster the relay trips.

Engineers choose different curve shapes based on the equipment being protected:

$$t(I) = TMS \left(\frac{k}{\left(\frac{I}{Is}\right)^{\alpha} - 1} \right)$$

Curve type	k	α
Standard inverse	0.140	0.020
Very inverse	13.5	1
Extremely inverse	80	2
Long time inverse	120	1

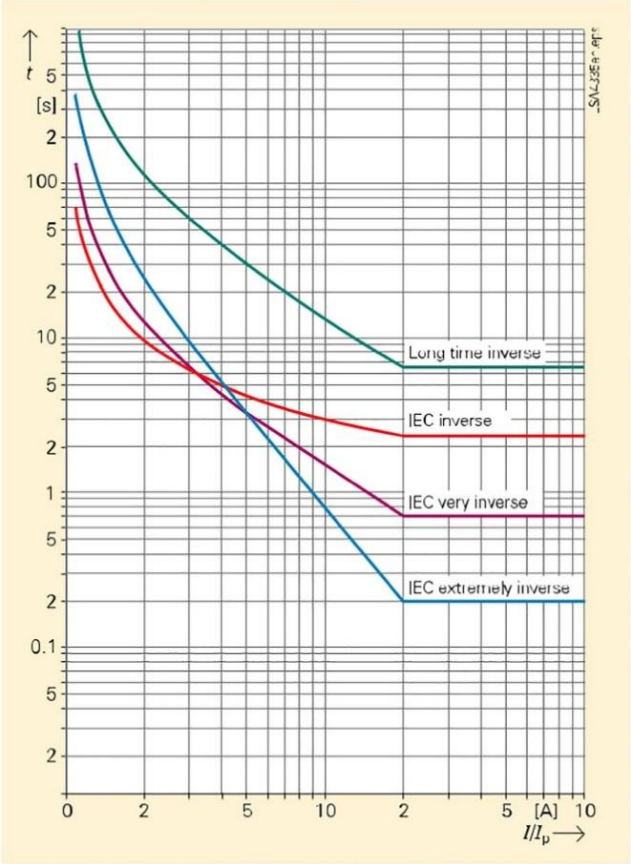


Figure 28 :- IDMT Curves

- Long Time Inverse: This curve is perfect for safeguarding equipment from persistent low-level overloads, like transformer thermal protection, because it has the slowest response time.

- IEC Inverse (Standard Inverse) : For general HV/MV protection, the most popular "all-rounder" configuration, which offers a minor variation in operating duration as the fault current increases.

- IEC Very Inverse: This curve's greater time-current slope makes it ideal for coordinating with fuses or situations where the fault current drastically decreases as you get farther away from the source.

- IEC Extremely Inverse: The steepest feature displayed, it offers extremely quick tripping for high-current short circuits and is intended to handle high-magnitude inrushes (such as motor starting).

5. Step-by-Step Calculation Guide

1. **Data Collection:** Map the system via a Single Line Diagram (SLD) and gather CT ratios, load data, and maximum fault levels.

2. **Curve Selection:** Pick the curve type (e.g., Standard Inverse, Very Inverse , Normal Inverse , etc) based on the application.

3. **Set the Pickup (Plug Setting):** This is usually set slightly above the maximum expected load (e.g., 120% of full load) so the relay doesn't trip under normal conditions.

$$I_s = \frac{\text{CT Primary Current}}{\text{CT Ratio}} \times \text{PS}$$

4. **Calculate Trip Time:** Use the standard IEC or ANSI formulas to see how the relay reacts to a specific fault current.

$$t = \frac{K \times \text{TMS}}{\left(\frac{I}{I_s}\right)^n - 1}$$

5. **Coordinate with TMS:** Start at the far end of the circuit with the lowest TMS. For every step moving back toward the power source, add a **coordination margin** (usually **0.2 to 0.4 seconds**) to ensure the downstream device always trips first.

   t upstream=t downstream+Δt

Where Δt = Coordination margin (typically 0.2 – 0.3 sec)

6. **Graphic Check:** Use software like ETAP to plot the TCC curves. If the curves touch or cross, the settings must be adjusted to prevent "nuisance tripping."

6. Typical Configuration Guidelines

Feeder Type	Typical Current Setting	Logic / Justification
Transformer	>120% of Full Load	Accounts for transformer capacity and downstream motor starts.
Motor	120-130% of Full Load	Must stay above normal running current but below damaging heat levels.
Incomer/Tie	110-120% of Max Load	Balanced to handle the total load of all connected downstream branches.

Table 2 :- Typical Settings Table

Chapter-10 Time-Current Characteristics

A **Time-Current Characteristic (TCC)** curve is a graphical representation used in power system protection for the analysis and coordination of protective devices such as relays, fuses, and circuit breakers. These curves are created and viewed using the Star-Systems Unified Analysis module in ETAP.

10.1 What is displayed by a TCC Curve?

Plotting on a log-log scale to accommodate a broad range of values, a TCC curve shows the relationship between:

1) X-axis (Current): Typically expressed as a multiple of the device's pickup current or in amperes.
2) Y-axis (Time): How long it takes the device to trip or "clear" the fault, often measured in seconds.

The working boundary of a device is seen when you examine a TCC curve in ETAP. The curve indicates the precise number of seconds that will elapse before that particular device interrupts the circuit in the event of a fault current.

10.2 What makes it significant?

Protective Device Coordination (PDC) is the main objective of employing TCC curves in ETAP.

It is essential for a number of reasons:

- **Selectivity (Coordination):** The device nearest to the issue should trip first. You can make sure upstream and downstream device curves don't overlap by superimposing them. This keeps the building's main breaker from tripping due to a minor issue in one room.
- **Equipment Protection:** "Damage Curves" for motors, transformers, and cables can all be plotted on the same graph. To guarantee that the breaker trips before the equipment is fried, the TCC curve of your breaker must be to the left and below the equipment damage curve.
- **Arc Flash Safety:** The "clearing time" of a fault, a crucial factor in determining Arc Flash hazard levels and necessary PPE, can be found using TCC curves.

- **Preventing Nuisance Tripping:** It aids in determining the "Inrush" margins for transformers and motors. You want to make sure that a motor's typical, extremely high beginning current doesn't surpass the TCC curve and result in an unintentional trip.

10.3 Important Elements of an ETAP TCC Plot

The following components are usually visible when you create a Star View in ETAP

1. **Long-Term/Pickup Delay:** The portion of the curve that deals with overloads (slow reaction).
2. **Instantaneous Trip:** For high-magnitude faults, the gadget trips instantly at the vertical drop at the bottom of the curve.
3. **Cording Gap:** The apparent "space" that separates two curves. To account for mechanical tolerances and relay error, engineers search for a gap that is large enough (often between 0.2 and 0.4 seconds).
4. **Motor Starting Curve:** Plotting the current profile as a motor spin up to make sure it doesn't "hit" the protection curve is known as a motor starting curve.
5.

1. Overview of the System

The protection study was conducted for the power distribution between the PCC 1 BUS and MCC-1 BUS. The following essential elements are part of the system:

- Incomer: ABB REF 615 relay protects PCC-1 I/C.
- Feeder: A 1/4/C 95 wire connects the outgoing (O/G) feeder to MCC-1.
- Loads: A lumped load of 0.425 MVA on the PCC-1 bus and 1.075 MVA on the MCC-1 bus.
- Power Factor Correction: Currently marked as "Open," a 500kvar capacitor bank (CAP3) is linked to the MCC-1 bus.

2. Protective Relay Settings

The coordination is achieved using **IEC Very Inverse** characteristics across all relays to ensure graded tripping.

Relay Coordination Table

Parameter	PCC-1 INCOMER (PCC-1 I/C R)	O/G TO MCC-1 (O/G TO MCC-1 R)	MCC-1 INCOMER (MCC-1 I/C R)
Relay Model	ABB REF 615	ABB REF 615	ABB REF 615
CT Ratio	3200:1	2000:1	2000:1
Pickup (A)	1.0 (0.05–5×CT Sec)	1.13 (0.05–5×CT Sec)	1.13 (0.05–5×CT Sec)
Time Dial	0.36	0.23	0.23
Inst. Setting	2.0× (1 - 40xCT Sec)	2.5× (1 - 40xCT Sec)	2.5× (1 - 40xCT Sec)
Inst. Delay	0.5 s	0.3 s	0.3 s

Table 3 :- Relay Settings of TCC Curve

The following is shown via the Time-Current Characteristic (TCC) plot:

- Operating Points: There is a noticeable coordination margin at three times pickup, with the PCC-1 Incomer operating at 2.43 s and the O/G feeder operating at 1.55 s.
- Cable Protection: The cable is shielded from thermal damage under fault circumstances by the "PCC TO MCC CABLE" curve, which is located to the right of the relay curves.
- Reference Voltage: A reference voltage of 0.436 kV is used to plot all curves.

Chapter-11 Arc Flash Analysis

11.1 IEEE 1584: Guide for Performing Arc-Flash Hazard Calculations

The IEEE 1584 standard provides the mathematical models and guidelines needed to calculate arc-flash incident energy and set arc-flash boundaries in three-phase electrical systems. The standard uses formulas derived from extensive lab tests to predict the heat generated by an arcing fault. These calculations are essential for an arc flash study. They help engineers assess the thermal hazard at a specific working distance and ensure that appropriate safety measures can be put in place for personnel near energized equipment.

11.2 NFPA 70E: Standard for Electrical Safety in the Workplace

While IEEE 1584 offers the mathematical framework for calculating arc flash hazards, NFPA 70E outlines how to apply those calculations to create safe work practices. This standard provides guidelines for choosing the right Personal Protective Equipment (PPE) based on the calculated incident energy. NFPA 70E requires that electrical equipment be de-energized before maintenance, unless de-energizing creates additional hazards or is impractical. When live work is needed, NFPA 70E demands clear labelling that shows the incident energy, arc flash boundary, and the required PPE category to protect workers from serious injuries.

11.3 Arc Flash Study Procedure

A thorough arc flash study follows a systematic approach, usually carried out with power system analysis software. The standard procedure includes these steps:

1. Data Collection: Gather accurate data, including one-line diagrams, equipment ratings, conductor lengths, and utility fault contribution.
2. Short Circuit Analysis: Calculate the maximum and minimum bolted three-phase fault currents at each bus to find the expected arcing current.
3. Protective Device Coordination Study: Analyse Time-Current Curves (TCC) to determine the clearing time of the upstream protective device during an arcing fault.
4. Arc Flash Hazard Calculations: Use IEEE 1584 formulas to calculate the incident energy and arc flash boundary at each equipment location.
5. Reporting and Labelling: Compile the results into a detailed report and create NFPA 70E-compliant hazard warning labels for equipment.

11.4 Arc Flash Terminology

To understand the arc flash study, several key terms need defining:

- Arc Flash: A dangerous energy release caused by an electrical fault that vaporizes metal and travels through the air from one conductor to another or to ground.
- Incident Energy: The amount of thermal energy absorbed on a surface, located at a specific working distance from the arc fault. It is typically measured in calories per square centimetre (cal/cm^2).
- Working Distance: The assumed distance from the arcing point to the face and chest of the worker. This distance can vary based on the type of equipment (e.g., 18 inches for a standard panel board).
- Arc Flash Boundary: The distance from a potential arc source within which a person could receive a second-degree burn if an electric arc flash happens.

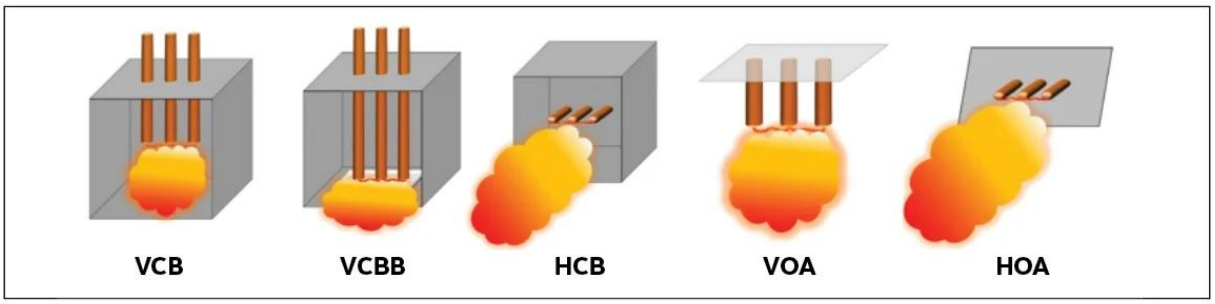
11.5 Input Required for Arc Flash Study

Precise calculations depend on accurate input data. The following parameters are essential for conducting the study:

11.5.1 Electrode Configuration

The physical arrangement of the conductors (electrodes) greatly affects how the arc plasma expands and directs thermal energy. According to the latest IEEE 1584 standards, common configurations include:

- **VCB** (Vertical Conductors in a Box): Arc energy is directed outward toward the worker.
- **VCBB** (Vertical Conductors terminated in an insulating Barrier): Similar to VCB but features an insulating barrier.
- **HCB** (Horizontal Conductors in a Box): Arc energy is projected directly at the worker, often resulting in the highest incident energy.
- **VOA** (Vertical Conductors in Open Air) and **HOA** (Horizontal Conductors in Open Air).



The arc flash direction in the various electrode/bus configurations

Figure 29 :- Different Electrodes Configurations

11.5.2 Gap Between Electrodes

The arc gap is the distance between the conductors or busbars where the arc occurs. A larger gap usually results in a higher arc voltage and greater incident energy. Gap sizes are generally standardized based on equipment class (e.g., 25 mm for low-voltage switchgear, 32 mm for motor control centres).

11.5.3 Existing Relay Setting

The incident energy is directly linked to how long the arc lasts. Therefore, the exact pickup and time-delay settings of the upstream protective relays or circuit breakers are crucial. Faster clearing times significantly reduce incident energy, making precise coordination data essential.

11.6 Understanding the Arc Flash Boundaries

- Arc Flash Boundary: This border represents a thermal hazard. It is the calculated distance 1.2 cal/cm^2 from an arc source at which unprotected skin would sustain a second-degree burn (beginning of harm). Wearing the proper Arc-Rated PPE is required for anyone crossing this barrier.
- Limited Approach Boundary: This boundary poses a risk of shock. For those who are not qualified, it is the furthest approach limit. Only when accompanied by a qualified individual and wearing the proper PPE can an unqualified individual cross this barrier. Its purpose is to avoid unintentional contact with components that are powered.

- Restricted Approach border: This is a shock hazard border located near the electrified conductor. It is only available to qualified individuals who use insulated tools and suitable shock PPE (such as rubber voltage-rated gloves) and typically require a documented work permit.

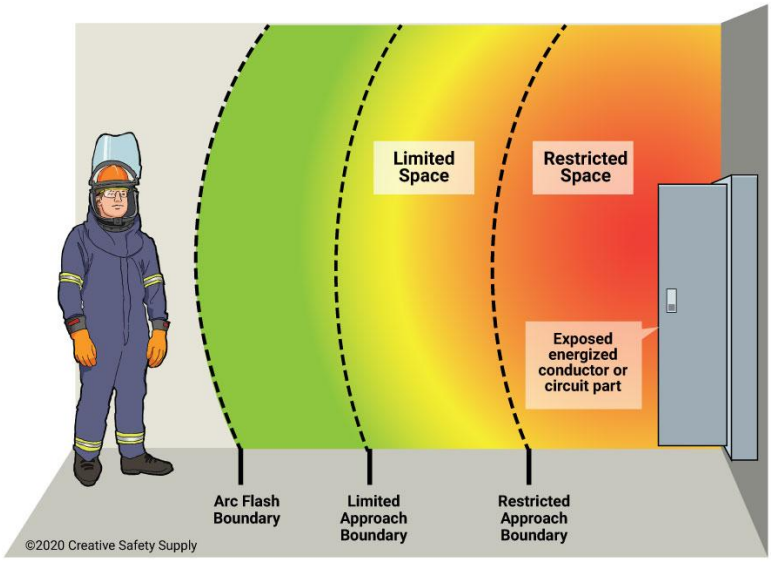


Figure 30 :- Arc Flash Boundaries

Nominal Potential Difference	Limited Approach Boundary		Restricted Approach Boundary; Includes Inadvertent Movement Adder
	Exposed Movable Conductor*	Exposed Fixed Circuit Part	
Less than 50 V	Not specified	Not specified	Not specified
50 V–300 V	3.0 m (10 ft. 0 in.)	1.0 m (3 ft. 6 in.)	Avoid contact
301 V–1 kV	3.0 m (10 ft. 0 in.)	1.0 m (3 ft. 6 in.)	0.3 m (1 ft. 0 in.)
1.1 kV–5 kV	3.0 m (10 ft. 0 in.)	1.5 m (5 ft. 0 in.)	0.5 m (1 ft. 5 in.)
5 kV–15 kV	3.0 m (10 ft. 0 in.)	1.5 m (5 ft. 0 in.)	0.7 m (2 ft. 2 in.)
15.1 kV–45 kV	3.0 m (10 ft. 0 in.)	2.5 m (8 ft. 0 in.)	0.8 m (2 ft. 9 in.)
45.1 kV–75 kV	3.0 m (10 ft. 0 in.)	2.5 m (8 ft. 0 in.)	1.0 m (3 ft. 6 in.)
75.1 kV–150 kV	3.3 m (10 ft. 8 in.)	3.0 m (10 ft. 0 in.)	1.2 m (3 ft. 10 in.)
150.1 kV–250 kV	3.6 m (11 ft. 8 in.)	3.6 m (11 ft. 8 in.)	1.6 m (5 ft. 3 in.)
250.1 kV–500 kV	6.0 m (20 ft. 0 in.)	6.0 m (20 ft. 0 in.)	3.5 m (11 ft. 6 in.)
500.1 kV–800 kV	8.0 m (26 ft. 0 in.)	8.0 m (26 ft. 0 in.)	5.0 m (16 ft. 5 in.)

Note: All dimensions are distance from exposed energized electrical conductors or circuit part to worker.
*Exposed movable conductors describes a condition in which the distance between the conductor and a person is not under the control of a person. The term is normally applied to overhead line conductors supported by poles.

SOURCE: NATIONAL FIRE PROTECTION ASSOCIATION

Figure 31 :- Arc Flash Boundary Marking (NFPA-70E)

11.7 Mitigation Techniques

When calculations show dangerously high incident energy levels, mitigation strategies must be adopted to protect personnel and equipment.

11.7.1. Arc Flash Detection Relay

What it is: This is a specialized protection relay designed to identify electrical arcs using optical sensors, often combined with overcurrent detection.
How it works: Traditional overcurrent relays wait to see if a current is from a harmless motor inrush or a short circuit. In contrast, Arc Flash Detection Relays use fibre-optic sensors or point sensors placed inside switchgear compartments, like bus compartments or breaker cells, to detect the unique burst of light generated by an arcing fault. When the relay sees this bright light and confirms it with a slight current rise, it sends a trip signal immediately to the upstream breaker.
In Arc Flash Analysis: This method significantly reduces arc duration ('t'). While a standard relay might take 0.3 seconds or longer to clear a bus fault due to coordination time delays, an arc flash relay can clear the fault in about 0.05 seconds (3-4 cycles), including breaker opening time. This greatly lowers the calculated incident energy and usually results in smaller PPE categories.

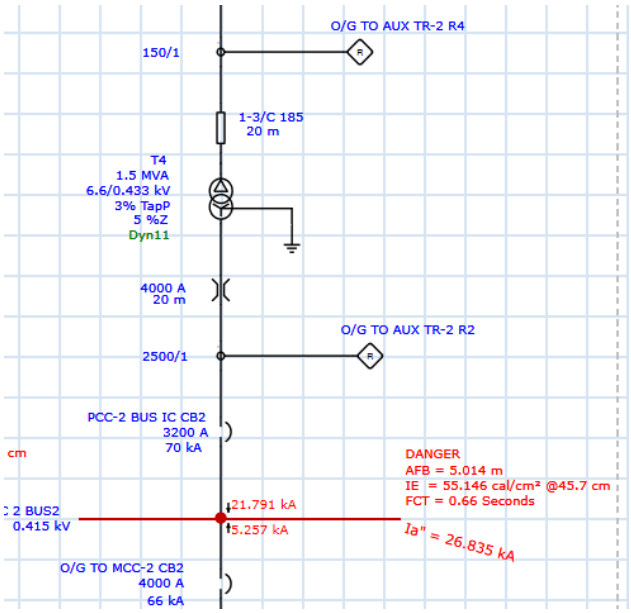


Figure 32 :- AFB Without Arc Flash Detection Relay

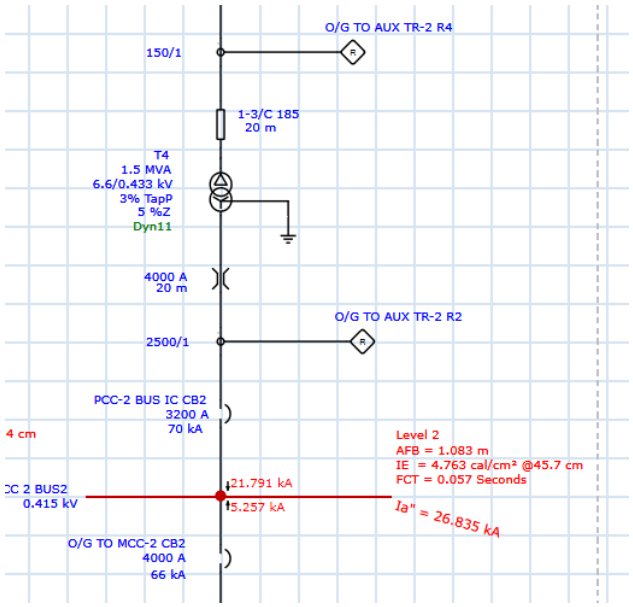


Figure 33 :- AFB With Arc Flash Detection Relay

The significant variation in Arc Flash Boundary (AFB) and Incident Energy (IE) based on the ETAP simulations shows how important protection speed is to power system safety.

Case 1: Standard Overcurrent Protection (DANGER)

The system uses standard Overcurrent (OC1 and OC2) stages in the "DANGER" scenario. In order to guarantee selectivity, these relays frequently have deliberate time delays. With a massive IE of 55.146 cal/cm^2, the Fault Clearing Time (FCT) in this case is 0.66 seconds.

This creates a deadly environment because it goes beyond the safe limit for the majority of PPE.

Case 2: Arc Detection Relay (Level 2)

The FCT is dramatically reduced to 0.057 seconds by using an Arc Detection Relay, which uses light sensors to detect the flash instantly. The energy fell to 4.763 cal/cm^2 because

$$IE \propto Time,$$

reducing the risk to a tolerable Level 2.

11.7.2. Zone Selective Interlocking (ZSI)

What it is: This is a communication scheme between upstream and downstream protective devices, like relays and modern circuit breaker trip units, that helps them locate faults without unnecessary time delays.
How it works: Consider a downstream breaker (Branch) and its upstream breaker (Feeder). When a fault happens below the downstream breaker, it quickly detects the fault and sends a "restrain" or "blocking" signal to the upstream breaker. The upstream breaker receives this signal and, knowing the fault is outside its zone, operates according to its standard coordination delay, allowing the downstream breaker to clear the fault.
In Arc Flash Analysis: The real benefit of ZSI shows when a fault occurs between the downstream and upstream breakers on the busbar. In this case, the upstream breaker sees the fault but does not receive a blocking signal from the downstream breaker. Knowing the fault is in its local zone, the upstream breaker overrides any intentional time delays and trips immediately to clear the bus fault. This allows for quick fault clearance on the main bus while keeping selectivity with downstream loads.

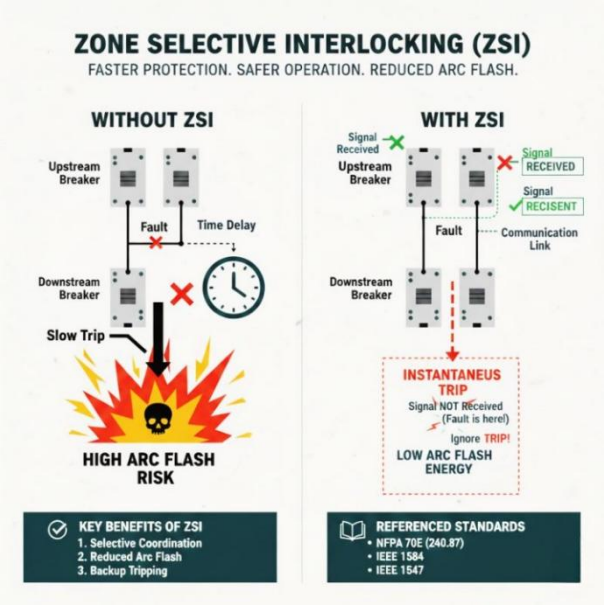


Figure 34 :- Zone Selective Interlocking

11.7.3. Energy Reducing Maintenance Switch (ERMS)

What it is: This is a physical switch or internal relay setting that temporarily changes the protective characteristics of a circuit breaker to its fastest setting whenever a qualified worker is present for maintenance.
How it works: When a qualified worker needs to work on or near energized equipment, like racking in a breaker or troubleshooting, they first flip the ERMS. This activates a pre-set maintenance mode in the breaker's digital trip unit, lowering the instantaneous pickup threshold and activating a 0-millisecond time delay setting. If an arc fault occurs while in maintenance mode, the breaker detects it almost immediately and trips instantly.
In Arc Flash Analysis: This is an operational mitigation technique. The analysis can show two levels of arc flash hazard for the equipment: one for normal operation and a second, much lower level for maintenance mode when the ERMS is active. The user must remember to flip the switch and reset it when finished, but this approach allows for temporary maintenance tasks to be done at lower PPE levels.

The crucial safety impact of an Energy Reducing Maintenance Switch (ERMS) is demonstrated by the ETAP simulations that are provided. Standard overcurrent relays (OC1/OC2) use deliberate time delays for coordination in the "DANGER" scenario. This results in a deadly 55.146 cal/cm^2 of Incident Energy (IE) and a slow Fault Clearing Time (FCT) of 0.66 seconds. The Arc Flash Boundary (AFB) reaches a perilous height of 5.014 meters.

On the other hand, an ERMS is activated during maintenance by the "Level 1" setup. By avoiding delays, the FCT is reduced to just 0.04 seconds. As a result, the risk is reduced to a manageable hazard with an AFB of 0.868 meters as the IE sharply decreases to 3.342 cal/cm^2.

11.7.4 Other Mitigation Techniques

Additional strategies include installing arc-resistant switchgear (designed to safely direct arc blast pressure and fire away from workers), using high-resistance grounding (HRG) to limit phase-to-ground fault currents, and utilizing remote racking devices that allow operators to insert or remove circuit breakers from outside the arc flash boundary.

11.8 Arc Flash Level

To simplify PPE selection, incident energy calculations are categorized into specific Arc Flash PPE Levels. Based on NFPA 70E, these levels range from Category 1 (requiring basic FR clothing with a rating of at least 4 cal/cm^2) to Category 4 (requiring a full arc flash suit with a rating of at least 40 cal/cm^2). If the calculated incident energy exceeds 40 cal/cm^2, it is

considered unsafe, and standard PPE is not adequate. Energized work is strictly prohibited until mitigation measures are put in place.

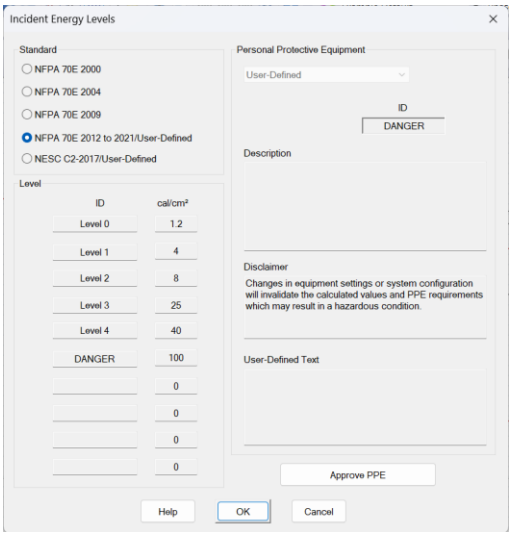


Figure 35 :- Level Marking With Incident Energy

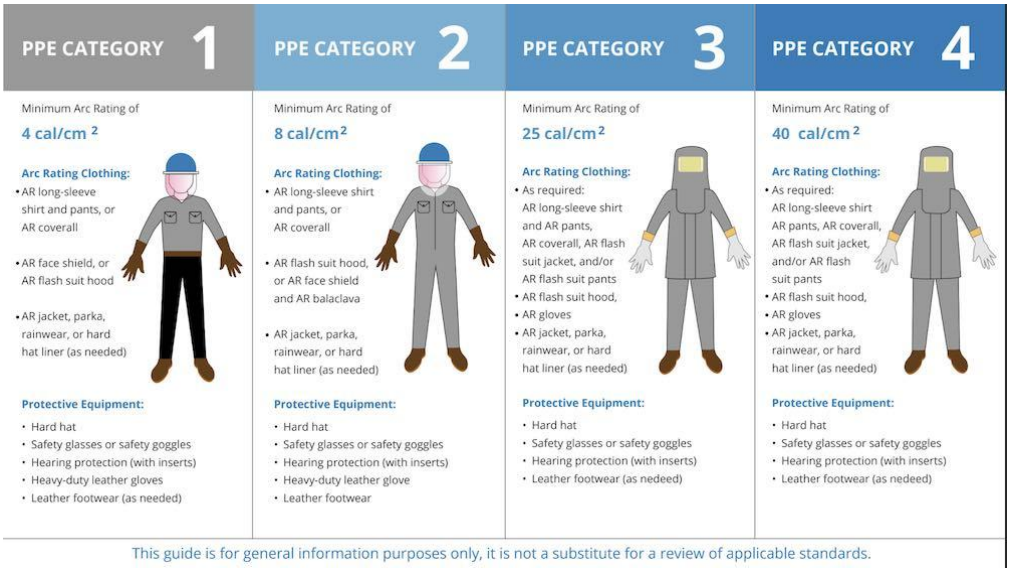


Figure 36 :- PPE Kits Classifications

Arc Flash Case Study

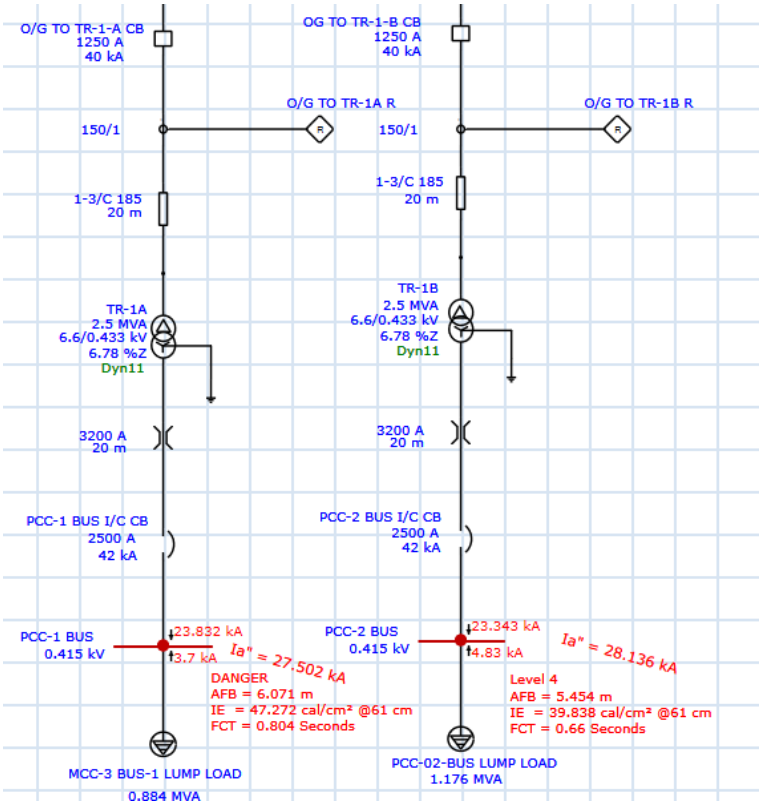


Figure 37 :- Case Study With OC1 and OC2 Settings Implementation

This is a classic instance of how arc flash hazards in power system studies are directly defined by protective relay settings. You are witnessing the direct, linear correlation between Arc Flash Incident Energy (IE) and Fault Clearing Time (FCT).This is a comprehensive clarification of how the arc flash hazard was reduced from "DANGER" to "Level 4," updated for a total Fault Clearing Time of 0.460 seconds, by adding that second stage to your ETAP relay settings.

The Fundamental Idea : Time is Energy

The incident energy is directly proportional to the fault duration, according to the fundamental principle of arc flash calculations (such as IEEE 1584). Less thermal energy is released when a relay instructs the breaker to open more quickly.

1. The PCC-1 BUS "DANGER" Feeder

The relay on your first feeder is set up with just one stage:

- **The setting** : The primary inverse-time overcurrent curve (OC1) is the only source of information.

- **The Reaction** : The relay follows that inverse curve when the arcing fault (I''_a = 27.502 kA) occurs. Its operation takes a considerable amount of time.
- **The Outcome** is a total Fault Clearing Time (FCT) of 0.804 seconds. Because of this extended duration, the incident energy can accumulate to 47.272 cal/cm^2, surpassing the 40 cal/cm^2 threshold and falling into the "DANGER" category (where standard PPE is typically deemed insufficient to protect a worker).

2. The PCC-2 BUS "Level 4" Feeder

By adding a second overcurrent stage (OC2) to the relay on your second feeder, you altered the regulations:

- **The Scene** : An OC2 stage with a pickup of 2x I_n (Full Load Current) and a time delay was added. In essence, this "clips" the slower inverse curve by acting as an instantaneous step or Definite Time Overcurrent (DTOC).
- **The Response** : The relay detects that the fault current greatly exceeds your 2x I_n threshold when the arcing fault happens on this bus (I''_a= 28.136 kA). As a result, it starts the fast OC2 timer right away and totally ignores the slow OC1 inverse curve.
- **As a result**, the relay runs considerably more quickly. This yields a total FCT of 0.460 seconds when combined with the breaker opening time.

Calculating the Effect of 0.460s FCT

Because the arcing current and the working distance remain constant, we can see exactly how your new 0.460s clearing time mathematically reduces the hazard by scaling the original ETAP calculation.

$$IE_{new} = IE_{original} \times \left(\frac{FCT_{new}}{FCT_{original}} \right)$$

- **Original Incident Energy:** 39.838 cal/cm^2
- **Original FCT:** 0.66 s
- **New FCT:** 0.460 s

$$IE_{new} = 39.838 \times \left(\frac{0.460}{0.66} \right)$$

$$IE_{new} \approx 27.76 \text{ cal/cm}^2$$

Conclusion

The project effectively created a digital twin and carried out a thorough electrical examination of the distribution system of a mineral processing plant using ETAP (Electrical Transient Analyzer Program) software. The study offered vital insights about the facility's operating performance, dependability, and safety by mimicking real-world settings. The thorough investigation produced a number of significant findings across four main engineering pillars:

- Load Flow Optimization: To guarantee stable voltage profiles at the 11 kV and 415 V Motor Control Centres (MCCs), the steady-state Load Flow Analysis assessed the system under peak output scenarios. This guarantees that heavy-duty equipment, including crushers and grinding mills, may effectively meet their high-power requirements without experiencing under-voltage tripping.
- System Resilience Against Faults: The maximum fault current magnitudes throughout the network were determined using Short Circuit Analysis, which was carried out in accordance with IEC 60909 standards. This confirmed that all switchgear and circuit breakers had sufficient interrupting ratings to manage fault conditions safely and avoid catastrophic equipment failures.
- Selective Protection Coordination: A selective protection strategy was created using Time-Current Characteristic (TCC) curves and relay coordination experiments. In order to isolate the problem while keeping the rest of the plant running and preventing expensive production downtime, this guarantees that protective devices are appropriately graded so that the device nearest to a fault trip first.
- Enhanced Personnel Safety: In compliance with IEEE 1584 and NFPA 70E standards, the Arc Flash Hazard Assessment meticulously determined incident energy levels and flash boundaries. According to the study, advanced mitigation techniques like Arc Flash Detection Relays, Zone Selective Interlocking (ZSI), and Energy Reducing Maintenance Switches (ERMS) can significantly shorten fault clearing times and reduce the necessary categories of Personal Protective Equipment (PPE) to safer levels.

In the end, this project's approaches and simulations provide a technical roadmap for the facility. The study effectively establishes the foundation for increased system stability, the avoidance of equipment failure, and the creation of a stringent, uniform safety protocol for all electrical workers by improving the electrical design and modernizing protective methods.

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